

SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

I. INTRODUCTION

1 Overview

1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:
<https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tracking-System>

1.2 Project team

Full Name	Role	Student ID	Email
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Bảng 1: Project Team Information

2 Project Scope and Limitations

2.1 Major Features

2.1.1 Major Features for User

- Register and log in to the system.
- Search scientific publications using keywords, authors, journals, or research fields.
- View detailed publication information.
- Save publications to bookmarks.

- Follow research topics of interest.
- Receive notifications about new publications and research trends.
- View publication trend analysis and statistical dashboards.
- Generate and export analytical reports (PDF/CSV).

2.1.2 Major Features for Administrator

- Log in and log out.
- Manage user accounts.
- Configure academic API sources (Semantic Scholar, OpenAlex, and Crossref).
- Synchronize publication data from external APIs.
- Monitor the data synchronization process.
- Manage system notifications.
- Generate system reports.
- Configure system settings.

2.2 Project Limitations

2.2.1 Limited Use of Full-Text Data

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

2.2.2 Restricted Data Scope

- The system focuses only on selected research fields (e.g., Computer Science and Artificial Intelligence).
- Therefore, it does not fully represent trends across all scientific disciplines.

2.2.3 Dependence on External APIs

- The system relies on external APIs such as Semantic Scholar, OpenAlex, and Crossref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

2.2.4 Simplified Trend Analysis and Update Frequency

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

II. PROJECT MANAGEMENT PLAN

1 Overview

1.1 Scope and estimation

#	WBS Item	Complexity	Duration (Days)
1. Project Initiation			
1.1	Define project scope	Medium	1
1.2	Collect and analyze requirements	Complex	2
2. Project Management and Environment Setup			
2.1	Set up development environment (GitHub, Database, Backend, Frontend)	Simple	1
2.2	Design database (ERD, Database Schema)	Complex	2
2.3	Configure security (JWT, RBAC, Password Hashing)	Complex	2
3. System Design			
3.1	Design Use Case Diagram and Activity Diagram	Medium	1
3.2	Design Sequence Diagram	Medium	1
3.3	Design DFD and ERD	Medium	1
4. System Development			
4.1	User Registration and Login	Medium	2
4.2	Search Scientific Publications	Complex	2
4.3	View Publication Details	Medium	1
4.4	Bookmark Management	Medium	1
4.5	Research Topic Following	Medium	1
4.6	Notification Management	Medium	1
4.7	Publication Trend Analysis	Complex	3
4.8	Develop Statistical Dashboard	Complex	2
4.9	Display Trending Topics	Medium	1
4.10	Generate Statistical Reports (PDF/CSV)	Complex	2

#	WBS Item	Complexity	Duration (Days)
5. Administration Functions			
5.1	User Management	Complex	2
5.2	Configure Academic API Sources	Complex	2
5.3	Synchronize Data from Semantic Scholar, OpenAlex, and Crossref	Complex	3
6. Testing			
6.1	Prepare Test Plan	Medium	1
6.2	Design Test Cases	Medium	2
6.3	Execute Testing	Complex	2
6.4	Fix Defects After Testing	Complex	2
7. Project Finalization			
7.1	Complete Project Documentation	Medium	2
7.2	Deploy the System	Medium	1
7.3	Prepare Presentation Slides	Simple	1
Total Duration			46 Working Days

1.2 Project Objective

- **Timeline:** The project must be completed before **July 3, 2026**.
- **Allocated Effort:** 46 man-days.

1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plan
1	External academic APIs (Semantic Scholar, OpenAlex, Crossref) become unavailable or change their endpoints.	The system cannot retrieve publication data, causing incomplete search results and inaccurate trend analysis.	High	Implement API fallback mechanisms, monitor API status regularly, and cache previously retrieved data whenever possible.

#	Risk Description	Impact	Possibility	Response Plan
2	Large volumes of publication data lead to slow system performance.	Users experience long response times when searching publications or viewing dashboards.	Medium	Optimize database indexing, implement pagination, caching, and asynchronous background synchronization.
3	Security vulnerabilities expose user accounts or stored data.	Unauthorized access, data leakage, and loss of user trust.	Medium	Use JWT authentication, RBAC authorization, password hashing, HTTPS, and regular security testing.
4	Trend analysis results become inaccurate because of outdated or incomplete datasets.	Generated reports and statistics may mislead users' research decisions.	Medium	Schedule automatic data synchronization and validate collected data before analysis.
5	System integration between frontend, backend, and database encounters compatibility issues.	Development delays and unexpected system failures during deployment.	Medium	Perform integration testing continuously and maintain version compatibility among all components.
6	Insufficient testing before deployment leaves critical defects unresolved.	System crashes, incorrect functionality, and poor user experience.	High	Prepare comprehensive Test Plans, Test Cases, conduct functional, integration, and user acceptance testing before release.
7	Project schedule delays due to underestimated implementation effort.	Failure to complete the project within the planned timeline.	Medium	Monitor project progress weekly, prioritize critical features, and adjust the implementation schedule when necessary.

2 Project Deliverables

Sprint	Week	Duration	Deliverables	Note
Sprint 1	Week 1	14/05 – 20/05/2026	• Define project scope • Analysis requirement	
Sprint 2	Week 2	21/05 – 27/05/2026	• Finish Product Overview • Finish User Requirement • Finish Functional Requirement	
Sprint 3	Week 3	28/05 – 03/06/2026	• Finish Database Design • Finish Software design description	
Sprint 4	Week 4	04/06 – 10/06/2026	• Finish Software requirement specification	
Sprint 5	Week 5	11/06 – 17/06/2026	• Finish Software design description (continued)	
Sprint 6	Week 6	18/06 – 24/06/2026	• Finish Software testing document • Finish Release package and user guide	
Sprint 7	Week 7	25/06 – 01/07/2026	• Testing • Finish document • Submitting	

3. Responsibility Assignment

Do ~D; Review ~R; Support ~S

Responsibility	Thanh Tú	Bảo Trâm	Huyền Trân	Trung Hiếu	Văn Học	Nhật Quang
Project Planning & Tracking	S	D	S	S	S	S
Prepare Project Introduction Document	R	D	R	R	R	R
Prepare Software Requirement Document	S	D	S	S	S	S
Prepare Software Design Document	D	D	D	D	D	D
Prepare Test Document	D	D	D	D	D	D
Prepare Software User Guides	D	D	D	D	D	D
Prepare Final Report	D	D	D	D	D	D

4. Project Communications

Communication Item	Who	Purpose	When	Type
Team Weekly Meeting	Group	- Report work project - Assign work for next week - Control project deadline	Every Sunday	Meeting on Google Meet
Weekly Report	Member	Report work process	1 day / week	Report via Zalo
Group Meeting	Group	Report work process	Every Friday	Offline on class

5. Configuration Management

5.1 Document Management

- Document tool: Confluence
- Task management tool: Jira

5.2 Source Code Management

- Source code management tool: GitHub

5.3 Tools & Infrastructure

Category	Tools / Infrastructure
Programming Languages	PHP, HTML5, CSS3, JavaScript
Database	MySQL
Database Administration	phpMyAdmin
Academic APIs	Semantic Scholar API, OpenAlex API, Crossref REST API
API Testing	Postman
Version Control	Git, GitHub
Project Management	Jira, Confluence
Diagramming	Draw.io, PlantUML
IDE / Code Editor	Visual Studio Code
Local Server	WAMP Server

III. SOFTWARE REQUIREMENT SPECIFICATION

1 Product Overview

The Scientific Journal Publication Trend Tracking System (SJPTTS) is a web-based application developed to help researchers, students, lecturers, and academic institutions search, monitor, and analyze scientific publication trends. The system integrates academic data from Semantic Scholar, OpenAlex, and Crossref to provide comprehensive and up-to-date publication information.

The system allows users to search scientific publications using keywords, authors, journals, research fields, or publication years. Users can view detailed publication information, bookmark publications, follow research topics, receive notifications about newly published papers, analyze publication trends through interactive dashboards, and export reports in PDF or CSV format.

Administrators can manage user accounts, configure academic API sources, synchronize publication data, monitor system operations, and maintain the overall performance of the platform. The system aims to provide an efficient solution for discovering scientific literature and supporting research through publication trend analysis.

2 User Requirements

2.1 Actors

2.1.1 User

User is the primary actor of the system. A user interacts with the system to access research-related functionalities such as searching papers, tracking publication trends, saving bookmarks, following topics, and receiving notifications.

Responsibilities:

- Access research-related features
- Search scientific publications
- Track research trends
- Manage bookmarks
- Follow research topics
- Receive notifications
- Generate reports

2.1.2 Student

Student is a specialized type of User. Students use the system to search for research papers, follow topics of interest, and monitor publication trends for academic learning and study purposes.

Responsibilities:

- Search research papers
- Save bookmarks
- Follow topics
- Analyze publication trends

2.1.3 Researcher

Researcher is a specialized type of User. Researchers use the system for research analysis, trend monitoring, and report generation to support academic and scientific activities.

Responsibilities:

- Search research papers
- Analyze research trends
- Generate analytical reports
- Monitor emerging research topics

2.1.4 Admin

Admin manages the overall system configuration, user management, and synchronization processes.

Responsibilities:

- Manage users
- Configure API sources
- Synchronize academic data
- Monitor system operations

2.1.5 Academic API

Academic API is an external system that provides publication metadata and research-related data.

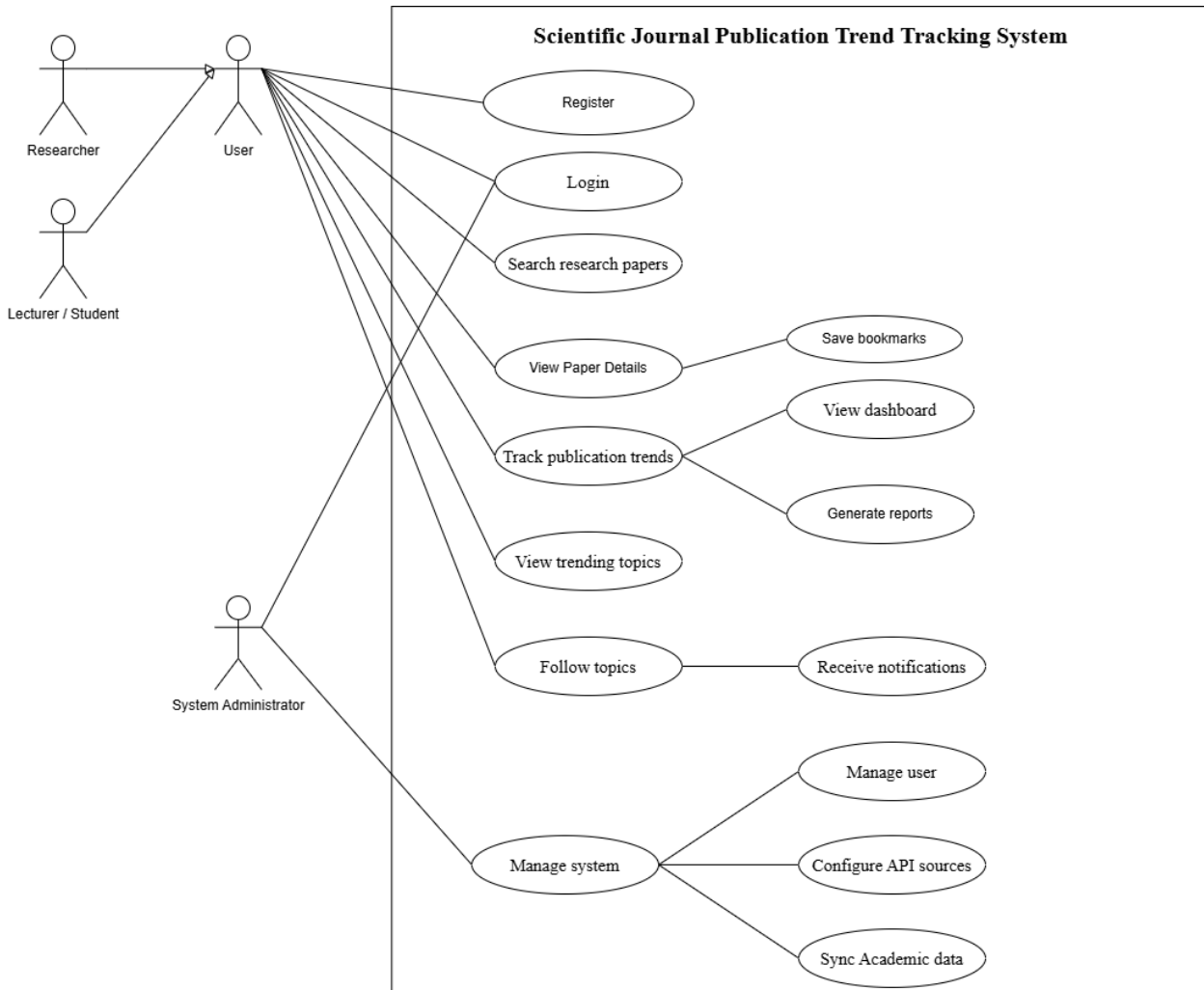
Examples:

- Semantic Scholar
- OpenAlex
- Crossref

Responsibilities:

- Provide publication metadata
- Support synchronization requests
- Return academic data to the system

2.2 Use Case Diagrams



Hình 1: Use Case Diagram

2.3 Use Case Descriptions

2.3.1 Register

Summary: This use case allows a new user to create an account in the Scientific Journal Publication Trend Tracking System. The user provides personal information and account credentials. The system validates the information, creates a new account, and stores the user data for future authentication and access to system functionalities.

Flow of Events

Main Flow

1. The user accesses the registration page.
2. The system displays the registration form.
3. The user enters the required information, including:

- Full Name
- Email Address
- Username
- Password

4. The user submits the registration form.
5. The system validates the entered information.
6. The system verifies that the username and email address are not already registered.
7. The system creates a new user account.
8. The system stores the account information in the database.
9. The system displays a registration success message.
10. The system prompts the user to proceed to the Login page.
11. The use case ends successfully.

Alternative Flow

Email Already Exists

1. The user submits the registration form.
2. The system detects that the email address is already registered.
3. The system displays: "This email address is already registered."
4. The user enters a different email address.
5. The main flow resumes from Step 4.

Username Already Exists

1. The user submits the registration form.
2. The system detects that the username already exists.
3. The system displays: "This username is already in use."
4. The user enters a different username.
5. The main flow resumes from Step 4.

Invalid Input Information

1. The system validates the submitted information.

2. One or more fields contain invalid data.
3. The system highlights the invalid fields.
4. The system displays validation error messages.
5. The user corrects the information.
6. The main flow resumes from Step 4.

Database Error

1. The system attempts to create the user account.
2. A database error occurs.
3. The system displays: “Unable to create account. Please try again later.”
4. The account is not created.
5. The use case ends.

Registration Cancelled

1. The user starts the registration process.
2. The user cancels the registration.
3. The system returns to the home page.
4. The use case ends.

Special Requirements

- Each email address must be unique.
- Each username must be unique.
- Passwords shall be stored securely using encryption or hashing mechanisms.
- Required fields shall be validated before account creation.
- Registration data shall be stored securely.
- The registration process should be completed within an acceptable response time.

Pre-condition

- The user does not already have an authenticated session.
- The registration service is available.

- The system database is operational.

Post-condition

- A new user account is successfully created.
- User information is stored in the database.
- The user is eligible to access the Login function.
- The system records the registration event.

Extension Point

Login

After successful registration, the user may proceed to log into the system.

2.3.2 Login

Summary: This use case allows users to authenticate themselves and access the Scientific Journal Publication Trend Tracking System according to their assigned roles.

Flow of Events

Main Flow

1. The user opens the system login page.
2. The user enters a valid username and password.
3. The user clicks the “Login” button.
4. The system verifies the user’s credentials.
5. The system identifies the user’s role.
6. The system grants access to the appropriate functions.
7. The system displays the main dashboard.

Alternative Flow

Invalid Credentials

1. The user enters an incorrect username or password.
2. The system displays an error message.
3. The user is prompted to re-enter login information.

Empty Input Fields

1. The user leaves the username or password field blank.

2. The system displays a validation message.
3. The user must complete the required fields before proceeding.

Special Requirements

- User passwords must be stored securely.
- Only authenticated users can access system functions.
- User sessions must be protected from unauthorized access.

Pre-condition

- The user has a registered account in the system.

Post-condition

- The user is successfully authenticated.
- A valid user session is created.
- The user gains access to authorized system functions.

Extension Point

None

2.3.3 Search Research Papers

Summary: This use case allows users to search for scientific publications using keywords, author names, or journal names. The system retrieves and displays publication metadata collected from synchronized academic data sources.

Flow of Events

Main Flow

1. The user accesses the Search Research Papers function.
2. The user enters one or more search criteria:
 - Keyword
 - Author name
 - Journal name
3. The user submits the search request.
4. The system validates the search criteria.
5. The system searches publication metadata stored in the system database.

6. The system retrieves matching records.
7. The system displays a list of matching publications, including:
 - Title
 - Authors
 - Journal
 - Publication Year
8. The user selects a publication from the result list.
9. The system invokes the View Paper Details use case.
10. The publication details are displayed to the user.

Alternative Flow

Empty Search Criteria

1. The user submits the search form without entering any criteria.
2. The system displays a validation message.
3. The user is requested to enter at least one search criterion.

No Matching Publications Found

1. The system performs the search.
2. No matching publications are found.
3. The system displays a “No results found” message.
4. The user may modify the search criteria and perform another search.

Data Retrieval Error

1. The system attempts to retrieve publication data.
2. The publication database is temporarily unavailable.
3. The system displays an error message.
4. The user is informed to try again later.

Special Requirements

- The system shall support searching by keyword, author, and journal.
- Search results should be returned within an acceptable response time.

- Search results should be ranked by relevance.
- Only publication metadata shall be displayed.

Pre-condition

- The user has successfully logged into the system.
- Publication metadata has been synchronized from external academic sources.

Post-condition

- A list of matching publications is displayed.
- The selected publication can be viewed in detail through the “View Paper Details” use case.

Extension Point

View Paper Details

After selecting a publication from the search results, the “View Paper Details” use case may be invoked to display detailed publication metadata.

2.3.4 View Paper Details

Summary: This use case allows users to view detailed information about a selected research paper, including metadata such as title, authors, abstract, keywords, publication year, and journal information. Users can review the paper before deciding to save it as a bookmark or use it for trend analysis.

Flow of Events

Main Flow

1. The user performs Search Research Papers.
2. The system displays a list of matching research papers.
3. The user selects a paper from the search results.
4. The system retrieves the paper metadata from the database or external academic API.
5. The system displays detailed information including:
 - Paper title
 - Authors
 - Abstract
 - Keywords

- Publication year
- Journal name
- Citation information (if available)

6. The user reviews the paper information.

7. The use case ends successfully.

Alternative Flow

Paper Information Not Available

1. The user selects a paper.
2. The system cannot retrieve the paper information.
3. The system displays an error message.
4. The user returns to the search results.

External API Unavailable

1. The system attempts to retrieve paper information from an external academic API.
2. The API does not respond or returns an error.
3. The system displays a notification indicating that paper details cannot be loaded.
4. The use case ends.

Save Bookmark

1. The user is viewing paper details.
2. The user chooses to save the paper.
3. The system invokes the Save Bookmarks use case.
4. The bookmark is stored successfully.

Special Requirements

- Paper details must be displayed within 3 seconds under normal conditions.
- The system shall display only metadata and not the full-text content of papers.
- Information retrieved from external APIs must be validated before display.
- The user interface should present paper information in a readable format.

Pre-condition

- The user has successfully logged into the system.
- The user has completed Search Research Papers and obtained search results.
- At least one paper exists in the result list.

Post-condition

- The selected paper's information is displayed.
- The user can continue to:
 - Save the paper as a bookmark
 - Return to search results
 - Continue exploring other papers

Extension Point

Save Bookmarks

The user may save the currently viewed paper to a personal bookmark list.

2.3.5 Save Bookmarks

Summary: This use case allows users to save research papers of interest to their personal bookmark collection for future reference and quick access.

Flow of Events

Main Flow

1. The user opens the details of a research paper through View Paper Details.
2. The user selects the Save Bookmark option.
3. The system verifies that the paper is not already bookmarked.
4. The system stores the paper information in the user's bookmark collection.
5. The system displays a confirmation message.
6. The use case ends successfully.

Alternative Flow

Paper Already Bookmarked

1. The user selects Save Bookmark.
2. The system detects that the paper already exists in the bookmark collection.

3. The system displays a notification: “This paper is already bookmarked.”
4. The use case ends.

User Session Expired

1. The user attempts to save a bookmark.
2. The system detects that the session has expired.
3. The system redirects the user to the Login page.
4. The use case ends.

Database Error

1. The system attempts to save the bookmark.
2. A database error occurs.
3. The system displays an error message.
4. The bookmark is not saved.
5. The use case ends.

Special Requirements

- A user cannot save duplicate bookmarks.
- Bookmark data must be associated with the correct user account.
- The system should confirm successful bookmark creation immediately.
- The bookmark operation should complete within 2 seconds under normal conditions.

Pre-condition

- The user has successfully logged into the system.
- The user has accessed “View Paper Details”.
- The selected paper exists in the system.

Post-condition

- The selected paper is stored in the user’s bookmark collection.
- The bookmark list is updated.
- The user can access the bookmarked paper later.

Extension Point

None

2.3.6 Track Publication Trends

Summary: This use case allows users to analyze publication trends based on keywords, research topics, authors, or journals. The system collects publication metadata from academic data sources and visualizes publication trends over time through charts and dashboards. This functionality helps researchers, lecturers, and students identify emerging research topics and understand the development of scientific fields.

Flow of Events

Main Flow

1. The user logs into the system.
2. The user selects “Track Publication Trends”.
3. The system displays a trend analysis interface.
4. The user enters one or more:
 - Keywords
 - Research topics
 - Author names
 - Journal names
5. The user specifies a time range for analysis.
6. The system retrieves related publication metadata from the database.
7. The system aggregates publication data by year.
8. The system analyzes publication frequency and growth patterns.
9. The system generates trend statistics.
10. The system invokes:
 - View Dashboard
 - View Trending Topics
11. The system displays visual charts and analytical results.
12. The user reviews the trend information.
13. The use case ends successfully.

Alternative Flow

No Matching Publications Found

1. The user submits a trend analysis request.
2. The system cannot find matching publications.
3. The system displays: “No publication data found for the selected criteria.”
4. The use case ends.

Invalid Time Range

1. The user selects an invalid time range.
2. The system validates the input.
3. The system displays an error message.
4. The user enters a valid time range.
5. The main flow resumes from Step 5.

Insufficient Data

1. The system retrieves only a small number of publications.
2. The system generates limited statistics.
3. The system displays: “Limited data available for trend analysis.”
4. The use case continues.

Data Synchronization Failure

1. The system attempts to access publication data.
2. Required data is unavailable.
3. The system displays an error message.
4. The use case ends.

Generate Report

1. The user reviews trend results.
2. The user selects “Generate Reports”.
3. The system invokes “Generate Reports”.
4. A report is generated based on the current trend analysis.

Special Requirements

- The system shall support trend analysis using:
 - Keywords
 - Topics
 - Authors
 - Journals
- The system shall visualize results using charts and dashboards.
- The system shall support analysis across multiple years.
- Trend calculations shall be based on publication metadata only.
- The system shall process trend queries within 5 seconds under normal conditions.
- The system shall support data collected from:
 - Semantic Scholar
 - OpenAlex
 - Crossref
- The system shall provide clear visual indicators of increasing or decreasing publication trends.

Pre-condition

- The user has successfully logged into the system.
- Publication metadata exists in the database.
- Academic data sources have been synchronized previously.
- The user has permission to access trend analysis features.

Post-condition

- Trend analysis results are generated.
- Publication statistics are displayed.
- Dashboard visualizations are available.
- Trending topics are identified.
- The user may generate reports based on the analysis.

Extension Point

View Dashboard

The system may display statistical dashboards showing publication counts and trend charts.

View Trending Topics

The system may identify and display currently popular research topics.

Generate Reports

The user may export or generate analytical reports from trend analysis results.

2.3.7 View Dashboard

Summary: This use case allows users to view dashboard statistics and visual representations of publication data. The dashboard provides an overview of publication trends, research activity, and topic popularity through charts and statistical indicators.

Flow of Events

Main Flow

1. The user accesses “Track Publication Trends”.
2. The system completes trend analysis.
3. The system invokes “View Dashboard”.
4. The system retrieves statistical data.
5. The system generates visual components, including:
 - Publication count by year
 - Topic distribution
 - Journal distribution
 - Growth trends
6. The system displays the dashboard.
7. The user reviews charts and statistics.
8. The use case ends successfully.

Alternative Flow

No Statistical Data Available

1. The system attempts to generate the dashboard.
2. No statistical data is available.

3. The system displays: “No statistical data available.”
4. The use case ends.

Visualization Generation Failure

1. The system attempts to generate charts.
2. An internal error occurs.
3. The system displays an error message.
4. The use case ends.

Special Requirements

- Dashboard information must be presented visually.
- Charts should be easy to understand.
- Statistical information must be updated according to synchronized publication data.
- Dashboard loading time should not exceed 5 seconds under normal conditions.

Pre-condition

- The user has successfully logged into the system.
- Trend analysis has been performed.
- Publication statistics are available.

Post-condition

- Statistical charts are displayed.
- Users gain an overview of publication trends.
- Users may continue to view trending topics or generate reports.

Extension Point

None

2.3.8 View Trending Topics

Summary: This use case allows users to identify research topics that are currently gaining attention based on publication trends and topic frequency analysis.

Flow of Events

Main Flow

1. The user accesses “Track Publication Trends”.
2. The system analyzes publication metadata.
3. The system identifies frequently occurring topics.
4. The system ranks topics according to popularity and growth rate.
5. The system displays a list of trending research topics.
6. For each topic, the system displays:
 - Topic name
 - Number of related publications
 - Trend direction
7. The user reviews the trending topics.
8. The use case ends successfully.

Alternative Flow

No Trending Topics Identified

1. The system performs trend analysis.
2. No significant topic trends are found.
3. The system displays: “No trending topics identified.”
4. The use case ends.

Insufficient Publication Data

1. The system analyzes available data.
2. The amount of data is insufficient.
3. The system displays a warning message.
4. The use case continues with limited results.

Special Requirements

- Topic ranking should be based on publication frequency and growth rate.
- Trending topics should be updated whenever publication data is synchronized.
- Results should be displayed clearly and intuitively.

Pre-condition

- The user has successfully logged into the system.
- Publication metadata exists in the database.
- Trend analysis has been performed.

Post-condition

- Trending research topics are displayed.
- Users can identify emerging research areas.
- Users may choose to follow topics for future updates.

Extension Point

Follow Topics

Users may choose to follow a trending topic for future notifications.

2.3.9 Follow Topics

Summary: This use case allows users to follow research topics of interest. The system records the user's preferences and monitors future publication activities related to those topics. Users can receive updates and notifications when new publications or significant trend changes occur.

Flow of Events

Main Flow

1. The user accesses the list of trending topics or trend analysis results.
2. The system displays available research topics.
3. The user selects a research topic.
4. The user clicks the Follow Topic button.
5. The system verifies that the topic is not already being followed by the user.
6. The system stores the topic in the user's followed-topic list.

7. The system confirms the follow action.
8. The use case ends successfully.

Alternative Flow

Topic Already Followed

1. The user selects a topic that is already followed.
2. The system detects the duplicate subscription.
3. The system displays: “You are already following this topic.”
4. The use case ends.

Invalid Topic Selection

1. The user selects an unavailable topic.
2. The system cannot locate the topic.
3. The system displays an error message.
4. The use case ends.

Database Storage Failure

1. The system attempts to save the followed topic.
2. A storage error occurs.
3. The system displays an error message.
4. The follow request is not completed.
5. The use case ends.

Special Requirements

- A user cannot follow the same topic multiple times.
- Followed topics must be associated with the correct user account.
- The system shall support following multiple topics simultaneously.
- Follow information must persist between user sessions.

Pre-condition

- The user has successfully logged into the system.

- The user has accessed trend analysis results or trending topics.
- The selected topic exists in the system.

Post-condition

- The selected topic is added to the user's followed-topic list.
- The system begins monitoring publication activities related to the topic.
- The user becomes eligible to receive notifications related to the topic.

Extension Point

Receive Notifications

The system may send notifications when new publications related to followed topics become available.

2.3.10 Receive Notifications

Summary: This use case allows users to receive notifications about newly published research papers, emerging research trends, and updates related to topics they follow. Notifications help users stay informed without manually checking publication databases.

Flow of Events

Main Flow

1. The user follows one or more research topics using "Follow Topics".
2. The system continuously monitors publication metadata during synchronization processes.
3. The system detects:
 - Newly published papers related to followed topics
 - Significant increases in publication activity
 - Emerging research trends
4. The system generates notification messages.
5. The system stores notifications in the user's notification center.
6. The user accesses the notification section.
7. The system displays available notifications.
8. The user selects a notification.
9. The system displays the related publication or trend information.

10. The use case ends successfully.

Alternative Flow

No New Notifications

1. The user accesses the notification center.
2. The system finds no new notifications.
3. The system displays: “No new notifications available.”
4. The use case ends.

Notification References Deleted Data

1. The user opens a notification.
2. The related publication information is unavailable.
3. The system displays: “The referenced publication is no longer available.”
4. The use case ends.

Notification Generation Failure

1. The system detects new publication activity.
2. A notification generation error occurs.
3. The system logs the error.
4. The notification is not delivered.
5. The use case ends.

Special Requirements

- Notifications shall only be generated for topics followed by the user.
- Notification information must be accurate and up-to-date.
- Duplicate notifications should be avoided.
- Notifications should remain accessible until viewed or removed by the user.
- The system shall support multiple notifications for different topics.

Pre-condition

- The user has successfully logged into the system.

- The user has followed at least one research topic.
- Publication metadata has been synchronized from external academic sources.

Post-condition

- Relevant notifications are generated and stored.
- Users are informed about new publications and trend updates.
- Users may access related publication details or trend information.

Extension Point

View Paper Details

The user may open a notification related to a newly published paper.

Track Publication Trends

The user may open a notification related to trend changes.

2.3.11 Generate Reports

Summary: This use case allows users to generate analytical reports based on publication trend analysis. The generated reports summarize publication statistics, trending topics, research growth patterns, and other relevant insights within a specified time period.

Flow of Events

Main Flow

1. The user completes Track Publication Trends.
2. The system displays trend analysis results.
3. The user selects “Generate Reports”.
4. The system displays report generation options.
5. The user specifies report parameters:
 - Research topic
 - Keyword
 - Journal
 - Time range
6. The user submits the report request.
7. The system retrieves relevant publication statistics.
8. The system compiles analytical information.

9. The system generates the report.
10. The system displays the generated report.
11. The user reviews the report.
12. The use case ends successfully.

Alternative Flow

No Data Available

1. The user requests report generation.
2. The system finds no matching publication data.
3. The system displays: “No data available for report generation.”
4. The use case ends.

Invalid Report Parameters

1. The user enters invalid report criteria.
2. The system validates the input.
3. The system displays an error message.
4. The user corrects the parameters.
5. The main flow resumes from Step 5.

Report Generation Failure

1. The system begins generating the report.
2. An internal processing error occurs.
3. The system displays an error message.
4. The report is not generated.
5. The use case ends.

Insufficient Data for Analysis

1. The system retrieves limited publication data.
2. The system generates a report with limited findings.
3. The system displays: “The report contains limited analytical results due to insufficient data.”

4. The use case continues.

Special Requirements

- Reports shall be generated from publication metadata only.
- Reports shall include statistical summaries and trend analysis results.
- Reports shall clearly present publication growth patterns.
- Report generation should complete within an acceptable response time.
- Generated reports should be readable and suitable for academic reference.

Pre-condition

- The user has successfully logged into the system.
- Trend analysis has been completed through Track Publication Trends.
- Relevant publication data exists in the system database.

Post-condition

- An analytical report is generated successfully.
- The report is displayed to the user.
- The report summarizes publication trends and research insights.

Extension Point

Track Publication Trends

This use case is initiated after trend analysis results become available.

2.3.12 Manage System

Summary: This use case allows the System Administrator to manage system operations, including user account management, configuration of academic data sources, and synchronization of publication metadata from external academic APIs. The administrator ensures that the system operates correctly and that publication data remains up-to-date.

Flow of Events

Main Flow

1. The System Administrator logs into the system.
2. The administrator accesses the Manage System function.
3. The system displays available administration options:

- Manage Users
- Configure API Sources
- Synchronize Academic Data

4. The administrator selects an administrative task.
5. The system executes the selected operation.
6. The system updates the corresponding configuration or data.
7. The system confirms successful completion.
8. The use case ends successfully.

Alternative Flow

Unauthorized Access

1. A non-administrator user attempts to access the “Manage System” function.
2. The system verifies user permissions.
3. The system denies access.
4. An authorization error message is displayed.
5. The use case ends.

Invalid Configuration

1. The administrator enters invalid API configuration information.
2. The system validates the configuration.
3. The system displays validation errors.
4. The administrator corrects the configuration.
5. The main flow resumes from Step 4.

Data Synchronization Failure

1. The administrator initiates data synchronization.
2. The external academic API is unavailable.
3. The synchronization process fails.
4. The system logs the error.

5. The system notifies the administrator.
6. The use case ends.

User Management Error

1. The administrator performs a user management operation.
2. The operation cannot be completed.
3. The system displays an error message.
4. No changes are applied.
5. The use case ends.

Special Requirements

- Only users with administrator privileges can access this functionality.
- All administrative actions shall be logged for auditing purposes.
- API configuration changes must be validated before being applied.
- Synchronization processes shall not affect system availability.
- The system shall support integration with:
 - Semantic Scholar API
 - OpenAlex API
 - Crossref API

Pre-condition

- The administrator has successfully logged into the system.
- The administrator account has valid administrative privileges.

Post-condition

- System settings are updated if modifications are made.
- User information is updated if user management operations are performed.
- Academic publication metadata is synchronized if synchronization is executed.
- Administrative actions are recorded in the system log.

Extension Point

Manage Users

The administrator may manage user accounts, roles, and permissions.

Configure API Sources

The administrator may configure external academic data sources used by the system.

Synchronize Academic Data

The administrator may initiate synchronization of publication metadata from external APIs.

2.3.13 Manage Users

Summary: This use case allows the System Administrator to manage user accounts within the Scientific Journal Publication Trend Tracking System. The administrator can view user information, update account details, activate accounts, deactivate accounts, and manage user roles and permissions. This functionality ensures proper access control and system security.

Flow of Events

Main Flow

1. The System Administrator logs into the system.
2. The administrator accesses the Manage Users function from the administration panel.
3. The system displays a list of registered users.
4. The administrator searches for or selects a specific user account.
5. The system displays detailed user information, including:
 - User ID
 - Username
 - Email address
 - Assigned role
 - Account status
 - Registration date
6. The administrator chooses a management operation:
 - View user
 - Update user information
 - Change user role
 - Activate account

- Deactivate account

7. The administrator submits the requested changes.
8. The system validates the submitted information.
9. The system updates the user account.
10. The system records the administrative action in the audit log.
11. The system displays a success confirmation message.
12. The use case ends successfully.

Alternative Flow

User Not Found

1. The administrator searches for a user.
2. The system cannot locate the specified account.
3. The system displays: “User account not found.”
4. The use case ends.

Invalid User Information

1. The administrator updates account information.
2. The system validates the input.
3. Invalid or incomplete information is detected.
4. The system displays validation errors.
5. The administrator corrects the information.
6. The main flow resumes from Step 7.

Role Assignment Conflict

1. The administrator attempts to assign an invalid role.
2. The system detects a role conflict.
3. The system displays an authorization error.
4. The role change is rejected.
5. The use case ends.

Database Update Failure

1. The system attempts to update user information.
2. A database error occurs.
3. The system displays an error message.
4. No changes are applied.
5. The use case ends.

Unauthorized Access

1. A non-administrator attempts to access Manage Users.
2. The system verifies permissions.
3. Access is denied.
4. The system displays: “Access denied. Administrator privileges required.”
5. The use case ends.

Special Requirements

- Only administrators may perform user management operations.
- All administrative actions shall be recorded in audit logs.
- User role modifications shall be validated before being applied.
- Account status changes shall take effect immediately.
- The system shall prevent unauthorized privilege escalation.

Pre-condition

- The administrator has successfully logged into the system.
- The administrator possesses valid administrative privileges.
- User accounts exist within the system.

Post-condition

- User information is updated successfully.
- User roles and permissions are modified if required.
- Account status changes are applied.

- Administrative actions are recorded in system logs.

Extension Point

Manage System

“Manage Users” is invoked as part of “Manage System”.

2.3.14 Configure API Sources

Summary: This use case allows the System Administrator to configure, update, test, and manage external academic data sources used by the Scientific Journal Publication Trend Tracking System. These sources provide publication metadata that supports paper searching, trend analysis, trending topic identification, and report generation. The administrator can manage API connections such as Semantic Scholar, OpenAlex, and Crossref to ensure reliable and accurate data acquisition.

Flow of Events

Main Flow

1. The System Administrator logs into the system.
2. The administrator accesses the Configure API Sources function.
3. The system displays a list of currently configured academic APIs.
4. The administrator selects one of the following actions:
 - Add API Source
 - Update API Configuration
 - Remove API Source
 - Test API Connection
5. The administrator enters or modifies API configuration information, including:
 - API Name
 - API Endpoint URL
 - Authentication Key (if required)
 - Synchronization Settings
6. The administrator submits the configuration.
7. The system validates the configuration parameters.
8. The system attempts to establish a connection with the specified API.
9. The API responds successfully.

10. The system saves the configuration settings.
11. The system records the action in the administration log.
12. The system displays a success confirmation message.
13. The use case ends successfully.

Alternative Flow

Invalid Configuration Data

1. The administrator enters incomplete or incorrect API information.
2. The system validates the submitted data.
3. Validation fails.
4. The system displays detailed validation errors.
5. The administrator corrects the information.
6. The main flow resumes from Step 6.

API Connection Failure

1. The system attempts to establish a connection.
2. The API server is unavailable or unreachable.
3. The connection test fails.
4. The system displays: “Unable to connect to the selected API source.”
5. The configuration is not activated.
6. The use case ends.

Authentication Failure

1. The system attempts to authenticate with the API.
2. The API key or authentication credentials are invalid.
3. The API rejects the request.
4. The system displays: “Authentication failed. Please verify API credentials.”
5. The use case ends.

Duplicate API Source

1. The administrator attempts to add an API source that already exists.
2. The system detects duplication.
3. The system displays: “This API source is already configured.”
4. The use case ends.

Database Storage Error

1. The system attempts to save API configuration settings.
2. A database error occurs.
3. The system displays an error message.
4. The configuration is not saved.
5. The use case ends.

Special Requirements

- Only administrators shall be allowed to configure API sources.
- API credentials must be stored securely.
- API endpoint validation must be performed before activation.
- All configuration changes must be logged.
- The system shall support integration with:
 - Semantic Scholar API
 - OpenAlex API
 - Crossref API
- Configuration updates should not interrupt existing system operations.

Pre-condition

- The administrator has successfully logged into the system.
- The administrator has valid administrative privileges.
- The system administration module is available.

Post-condition

- API source configurations are successfully stored.

- New or updated API sources become available for synchronization.
- Configuration changes are recorded in the system log.
- The system is prepared to retrieve publication metadata from configured sources.

Extension Point

Synchronize Academic Data

After a successful API configuration, the administrator may initiate data synchronization.

2.3.15 Synchronize Academic Data

Summary: This use case allows the System Administrator to synchronize publication meta-data from external academic data sources into the system database. The synchronization process retrieves updated publication information and ensures that the system contains the latest data required for paper searching, trend analysis, dashboard visualization, trending topic identification, and report generation.

The system supports synchronization with external academic APIs such as Semantic Scholar, OpenAlex, and Crossref.

Flow of Events

Main Flow

1. The System Administrator logs into the system.
2. The administrator accesses the Synchronize Academic Data function.
3. The system displays available academic data sources.
4. The administrator selects one or more configured API sources.
5. The administrator initiates the synchronization process.
6. The system establishes connections to the selected APIs.
7. The system sends data retrieval requests.
8. The external APIs return publication metadata.
9. The system validates the received data.
10. The system identifies:
 - New publications
 - Updated publication records
 - Existing records

11. The system processes the retrieved metadata.
12. The system inserts new publication records into the database.
13. The system updates modified publication records.
14. The system removes duplicate records.
15. The system records synchronization statistics, including:
 - Number of records retrieved
 - Number of new records added
 - Number of updated records
 - Number of duplicate records detected
16. The system stores synchronization logs.
17. The system displays a synchronization summary report.
18. The use case ends successfully.

Alternative Flow

API Source Unavailable

1. The system attempts to connect to an academic API.
2. The API server is unavailable.
3. The connection request fails.
4. The system records the error.
5. The administrator is notified.
6. The synchronization process continues with other available sources.
7. The use case continues.

Network Connection Failure

1. The synchronization process begins.
2. A network interruption occurs.
3. The system cannot retrieve data.
4. The system displays: "Synchronization failed due to network connectivity issues."

5. The synchronization process stops.
6. The use case ends.

Invalid Retrieved Data

1. The system receives incomplete or malformed metadata.
2. The system validates incoming records.
3. Invalid records are rejected.
4. The system logs validation errors.
5. Valid records continue to be processed.
6. The use case resumes from Step 10.

Duplicate Publication Records

1. The system retrieves metadata already stored in the database.
2. The system identifies duplicate records.
3. Duplicate entries are ignored or merged.
4. The synchronization process continues.

Database Update Failure

1. The system attempts to store synchronized data.
2. A database error occurs.
3. The system logs the error.
4. The administrator is notified.
5. The synchronization process is terminated.
6. The use case ends.

No New Data Available

1. The system completes synchronization.
2. No new publication metadata is found.
3. The system displays: “No new publication data available.”
4. The synchronization summary is generated.

5. The use case ends successfully.

Special Requirements

- Only publication metadata shall be synchronized.
- Full-text articles or PDF files shall not be stored.
- Synchronization operations shall be logged for auditing purposes.
- Duplicate publication records shall be automatically detected and handled.
- The synchronization process shall support:
 - Semantic Scholar API
 - OpenAlex API
 - Crossref API
- Synchronization should be executable on a scheduled basis (daily or weekly).
- The synchronization process shall not affect normal user operations.

Pre-condition

- The administrator has successfully logged into the system.
- Academic API sources have been configured successfully.
- Network connectivity is available.
- External academic APIs are accessible.

Post-condition

- Publication metadata is updated in the system database.
- New publications become available for searching.
- Trend analysis data is refreshed.
- Dashboard statistics are updated.
- Trending topic calculations can use the latest data.
- Synchronization logs are stored for future reference.

Extension Point

Search Research Papers

After synchronization, users can search newly imported publications.

Track Publication Trends

Updated metadata can be used for trend analysis.

View Trending Topics

Newly synchronized data may affect topic popularity rankings.

2.4 Functional Requirements

2.4.1 User Authentication and Authorization

- Users can register a new account.
- Users can log in using a valid username and password.
- The system assigns permissions based on user roles.
- Administrators can manage user accounts and roles.

2.4.2 Research Paper Search

- Users can search research papers by keyword, author name, or journal name.
- The system displays matching publication results.
- Users can view detailed publication information.

2.4.3 Bookmark Management

- Users can save research papers to personal bookmarks.
- The system prevents duplicate bookmarks.
- Users can access bookmarked papers later.

2.4.4 Publication Trend Analysis

- Users can analyze publication trends based on keywords, topics, authors, or journals.
- The system generates trend statistics from publication metadata.
- The system visualizes trends using charts and dashboards.

2.4.5 Dashboard Visualization

- Users can view publication statistics through dashboards.
- The dashboard displays publication counts, topic distributions, journal distributions, and growth trends.

2.4.6 Trending Topics

- Users can view trending research topics.
- The system ranks topics according to publication frequency and growth rate.

2.4.7 Topic Following

- Users can follow research topics of interest.
- The system stores followed topics for each user.
- Users may follow multiple topics.

2.4.8 Notification Management

- Users receive notifications related to followed topics.
- Notifications include new publications, trend changes, and emerging topics.
- Users can view notification details.

2.4.9 Report Generation

- Users can generate analytical reports based on publication trends.
- Reports summarize publication statistics and research trends.
- Reports are generated from publication metadata.

2.4.10 System Administration

- Administrators can manage system operations.
- Administrators can manage user accounts.
- Administrators can configure academic API sources.
- Administrators can synchronize publication metadata from external APIs.

2.4.11 Academic Data Synchronization

- The system synchronizes publication metadata from external academic APIs.
- The system validates retrieved metadata.
- The system inserts new records and updates existing records.
- The system records synchronization logs and statistics.

2.5 Non-Functional Requirements

2.5.1 Usability

The interface is intuitive and accessible for all user types.

2.5.2 Performance

Search results must be returned within 2 seconds.

The system must support at least 500 concurrent users.

2.5.3 Reliability

System uptime must be at least 99% during business hours.

Data must be backed up daily.

2.5.4 Security

User passwords must be hashed before storage.

Role-based access control must be enforced for all features.

2.5.5 Maintainability & Scalability

The system must use a modular architecture for easy maintenance.

New data sources can be added without redesigning the existing system.

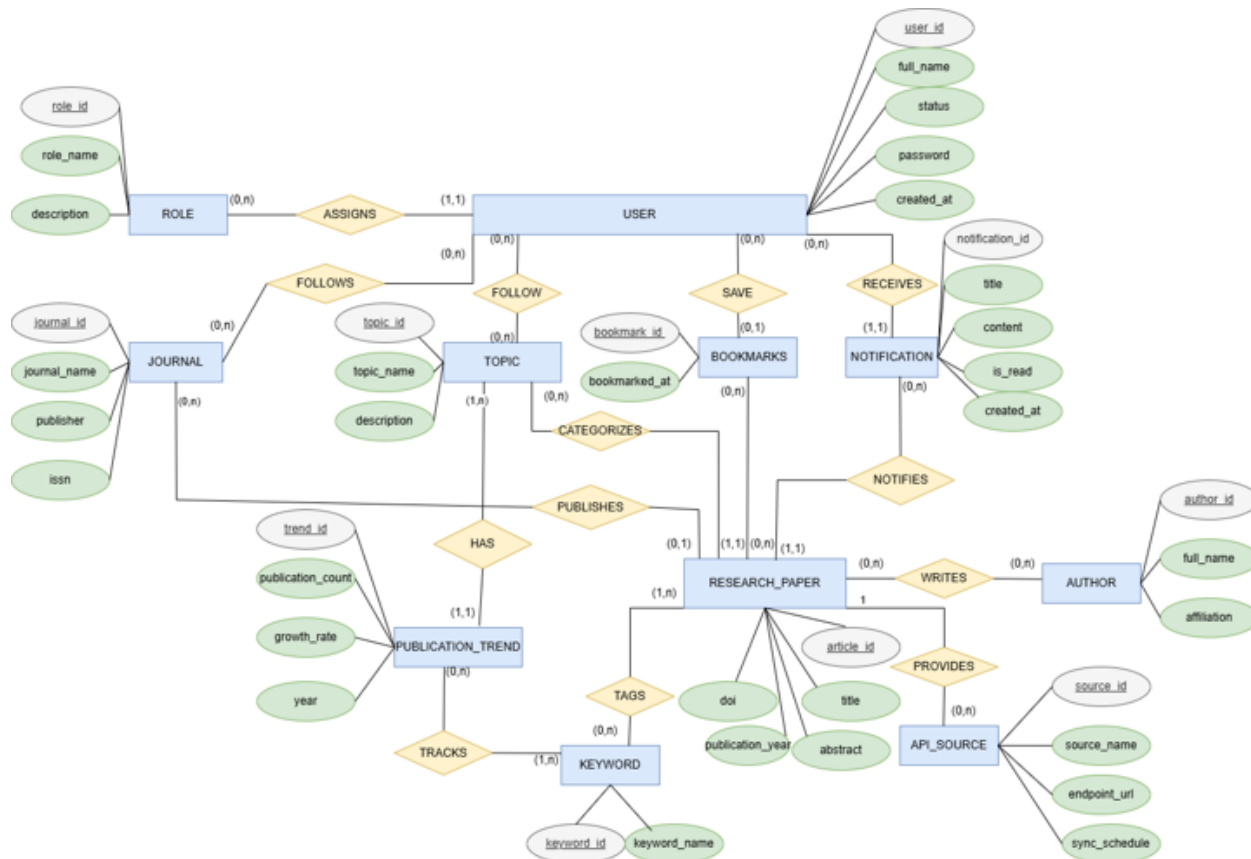
2.6 Screen Authorization

Screen	Researcher	Lecturer/Stude	System Administrator
Login	x	x	x
Register	x	x	
Search Research Papers	x	x	
View Paper Details	x	x	
Save Bookmarks	x	x	
Track Publication Trends	x	x	
View Dashboard	x	x	
Generate Reports	x	x	
View Trending Topics	x	x	
Follow Topics	x	x	
Receive Notifications	x	x	
Manage System			x
Manage Users			x
Configure API Sources			x
Sync Academic Data			x

2.7 Non-Screen Functions

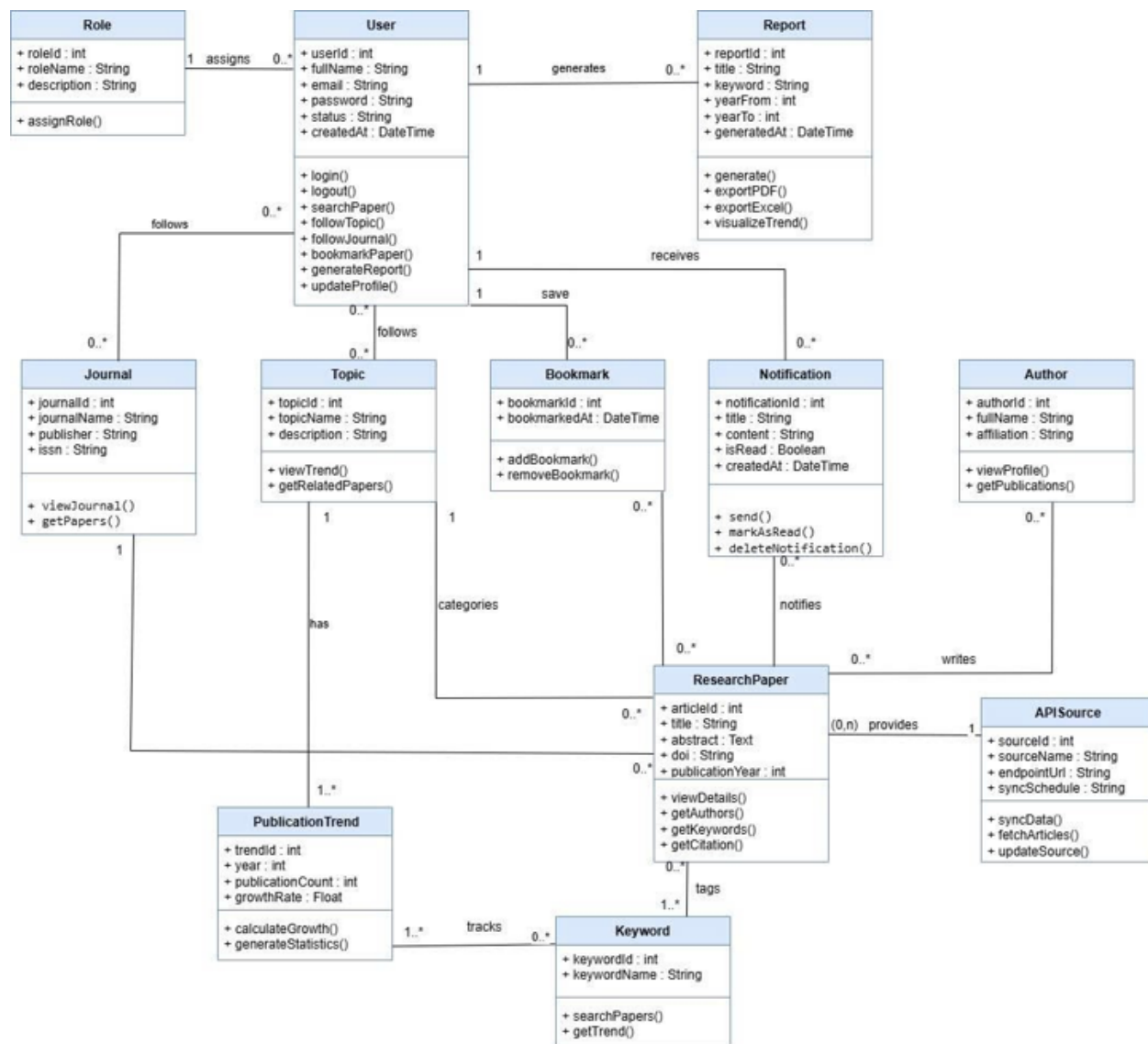
#	Feature	System Function	Description
1	Automated Academic Data Synchronization	Scheduled Job	Scheduled job that automatically retrieves publication metadata from external academic APIs (Semantic Scholar, OpenAlex, Crossref). Inserts new records, updates existing ones, removes duplicates, and logs synchronization statistics in the database.
2	Automated Notification Generation	Scheduled Job	Scheduled job that monitors followed topics for each user. When new publications related to followed topics are detected after synchronization, the system automatically generates and stores notifications in the database for the relevant users.
3	Automated Trend Calculation	Scheduled Job	Scheduled job that recalculates publication trend statistics after each data synchronization. Updates keyword frequency, topic growth rate, and journal distribution data used by the dashboard and trending topics features.

2.8 Entity Relationship Diagram



Hình 2: Entity Relationship Diagram (ERD)

2.9 Class Diagram



Hình 3: Class Diagram

IV. Software Design Description

1 System Design

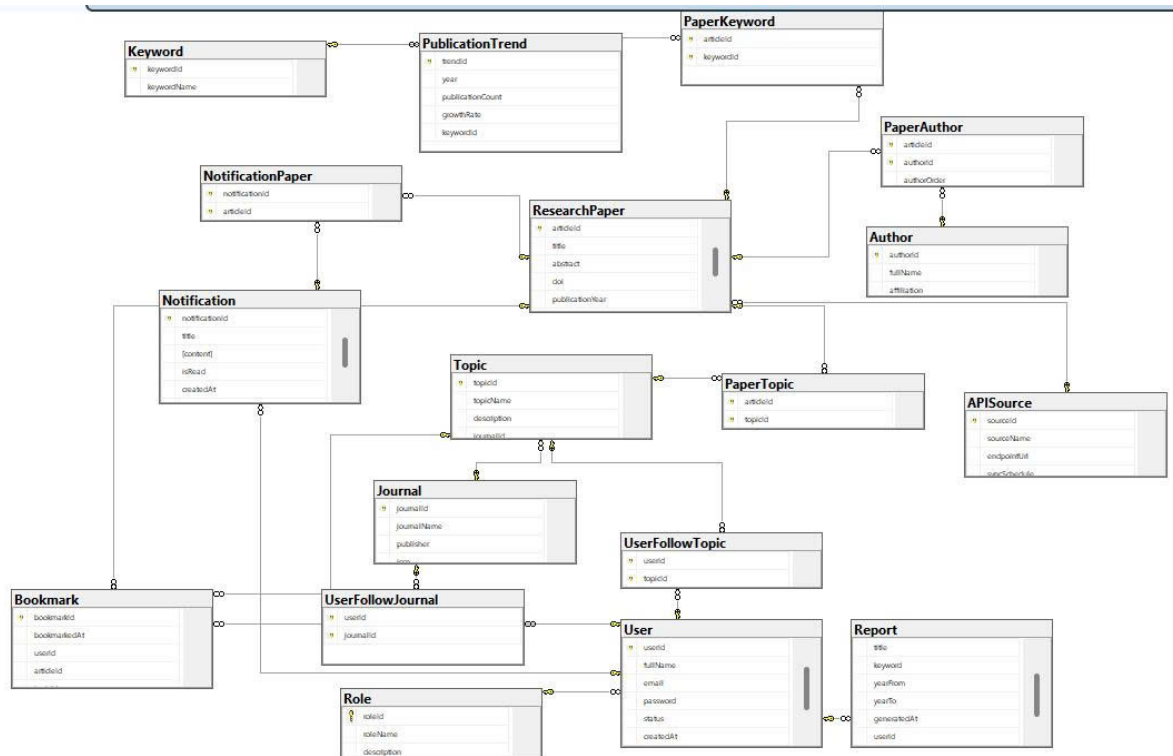
1.1 System Architecture

1.2 Package Diagram

1.2.1 Front End

1.2.2 Back End

2 Database Design



Hình 4: Database Design

Entities Description

2.1 Role Table

Field	Type	Description	PK	FK
roleId	INT	Role ID	Yes	No
roleName	NVARCHAR(100)	Role name	No	No
description	NVARCHAR(255)	Description	No	No

2.2 User Table

Field	Type	Description	PK	FK
userId	INT	User ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
email	NVARCHAR(150)	Email	No	No
password	NVARCHAR(255)	Password	No	No
status	NVARCHAR(50)	Status	No	No
createdAt	DATETIME2	Created time	No	No
roleId	INT	Role reference	No	Yes → Role

2.3 Journal Table

Field	Type	Description	PK	FK
journalId	INT	Journal ID	Yes	No
journalName	NVARCHAR(200)	Journal name	No	No
publisher	NVARCHAR(150)	Publisher	No	No
issn	NVARCHAR(20)	ISSN	No	No

2.4 UserFollowJournal (M:N)

Field	Type	Description	PK	FK
-------	------	-------------	----	----

userId	INT	User	Yes (Composite)	Yes → User
journalId	INT	Journal	Yes (Composite)	Yes → Journal

2.5 Topic Table

Field	Type	Description	PK	FK
topicId	INT	Topic ID	Yes	No
topicName	NVARCHAR(150)	Topic name	No	No
description	NVARCHAR(MAX)	Description	No	No
journalId	INT	Journal reference	No	Yes → Journal

2.6 UserFollowTopic (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Composite)	Yes → User
topicId	INT	Topic	Yes (Composite)	Yes → Topic

2.7 APISource Table

Field	Type	Description	PK	FK
sourceId	INT	Source ID	Yes	No
sourceName	NVARCHAR(150)	Source name	No	No
endpointUrl	NVARCHAR(500)	API URL	No	No
syncSchedule	NVARCHAR(100)	Sync schedule	No	No

2.8 Author Table

Field	Type	Description	PK	FK
authorId	INT	Author ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
affiliation	NVARCHAR(255)	Affiliation	No	No

2.9 Keyword Table

Field	Type	Description	PK	FK
keywordId	INT	Keyword ID	Yes	No
keywordName	NVARCHAR(100)	Keyword	No	No

2.10 Research Paper Table

Field	Type	Description	PK	FK
articleId	INT	Paper ID	Yes	No
title	NVARCHAR(500)	Title	No	No
abstract	NVARCHAR(MAX)	Abstract	No	No
doi	NVARCHAR(200)	DOI	No	No
publicationYear	INT	Year	No	No
sourceId	INT	API Source	No	Yes → APISource

2.11 PaperTopic (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → Research-Paper

topicId	INT	Topic	Yes (Composite)	Yes → Topic
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2.12 PaperKeyword (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → Research-Paper
keywordId	INT	Keyword	Yes (Composite)	Yes → Keyword

2.13 PaperAuthor (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → Research-Paper
authorId	INT	Author	Yes (Composite)	Yes → Author
authorOrder	INT	Author order	No	No

2.14 PublicationTrend Table

Field	Type	Description	PK	FK
trendId	INT	Trend ID	Yes	No
year	INT	Year	No	No
publicationCount	INT	Count	No	No
growthRate	FLOAT	Growth rate	No	No

keywordId	INT	Keyword	No	Yes → Key- word
-----------	-----	---------	----	-----------------------

2.15 Report Table

Field	Type	Description	PK	FK
reportId	INT	Report ID	Yes	No
title	NVARCHAR(300)	Title	No	No
keyword	NVARCHAR(200)	Keyword	No	No
yearFrom	INT	From year	No	No
yearTo	INT	To year	No	No
generatedAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.16 Notification Table

Field	Type	Description	PK	FK
notificationId	INT	Notification ID	Yes	No
title	NVARCHAR(300)	Title	No	No
content	NVARCHAR(MAX)	Content	No	No
isRead	BIT	Read status	No	No
createdAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.17 NotificationPaper (M:N)

Field	Type	Description	PK	FK
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notificationId	INT	Notification	Yes (Composite)	Yes → Notifi- cation
articleId	INT	Paper	Yes (Composite)	Yes → Re- search- Paper

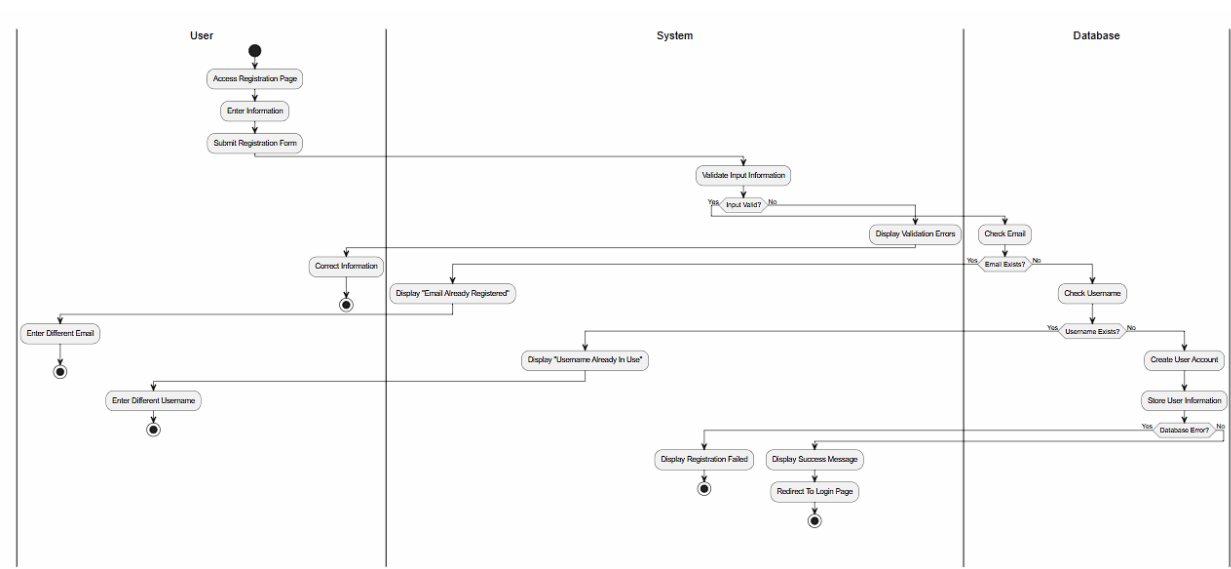
2.18 Bookmark Table

Field	Type	Description	PK	FK
bookmarkId	INT	Bookmark ID	Yes	No
bookmarkedAt	DATETIME2	Time saved	No	No
userId	INT	User	No	Yes → User
articleId	INT	Paper	No	Yes → Re- search- Paper
topicId	INT	Topic	No	Yes → Topic

3 Detail Design

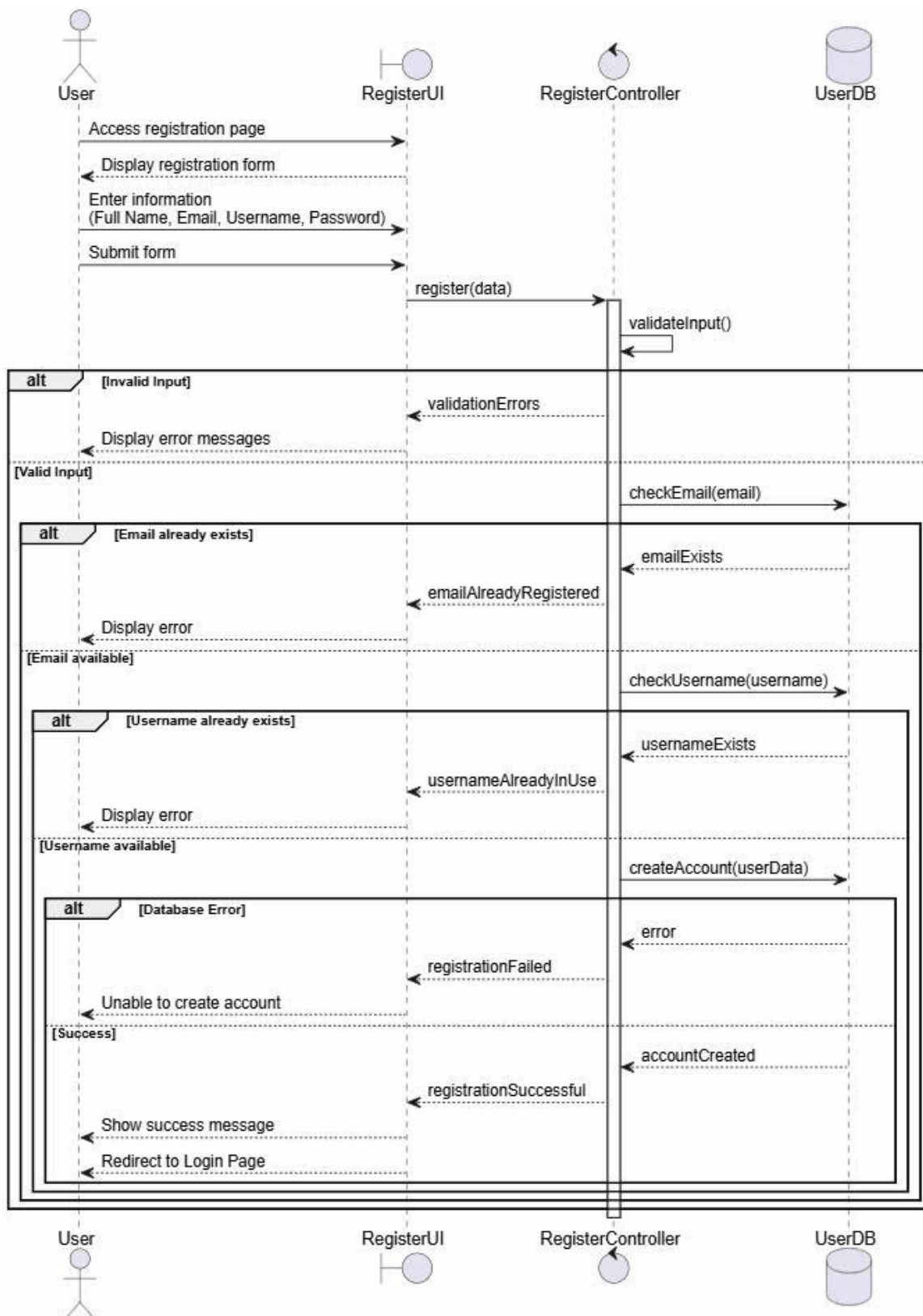
3.1 Register

3.1.1 Activity diagram



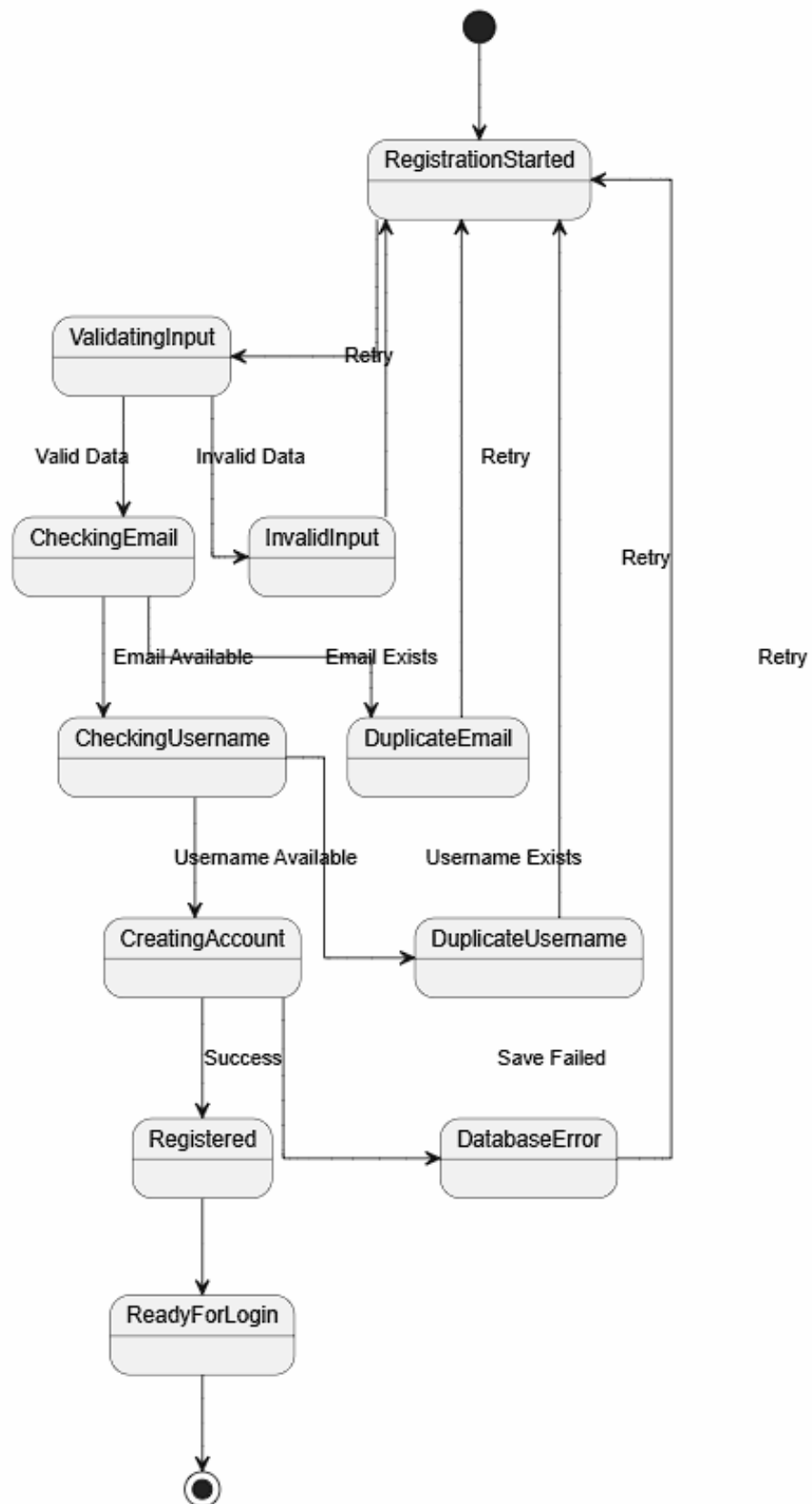
Hình 5: Activity diagram

3.1.2 Sequence Diagram



Hình 6: Sequence Diagram

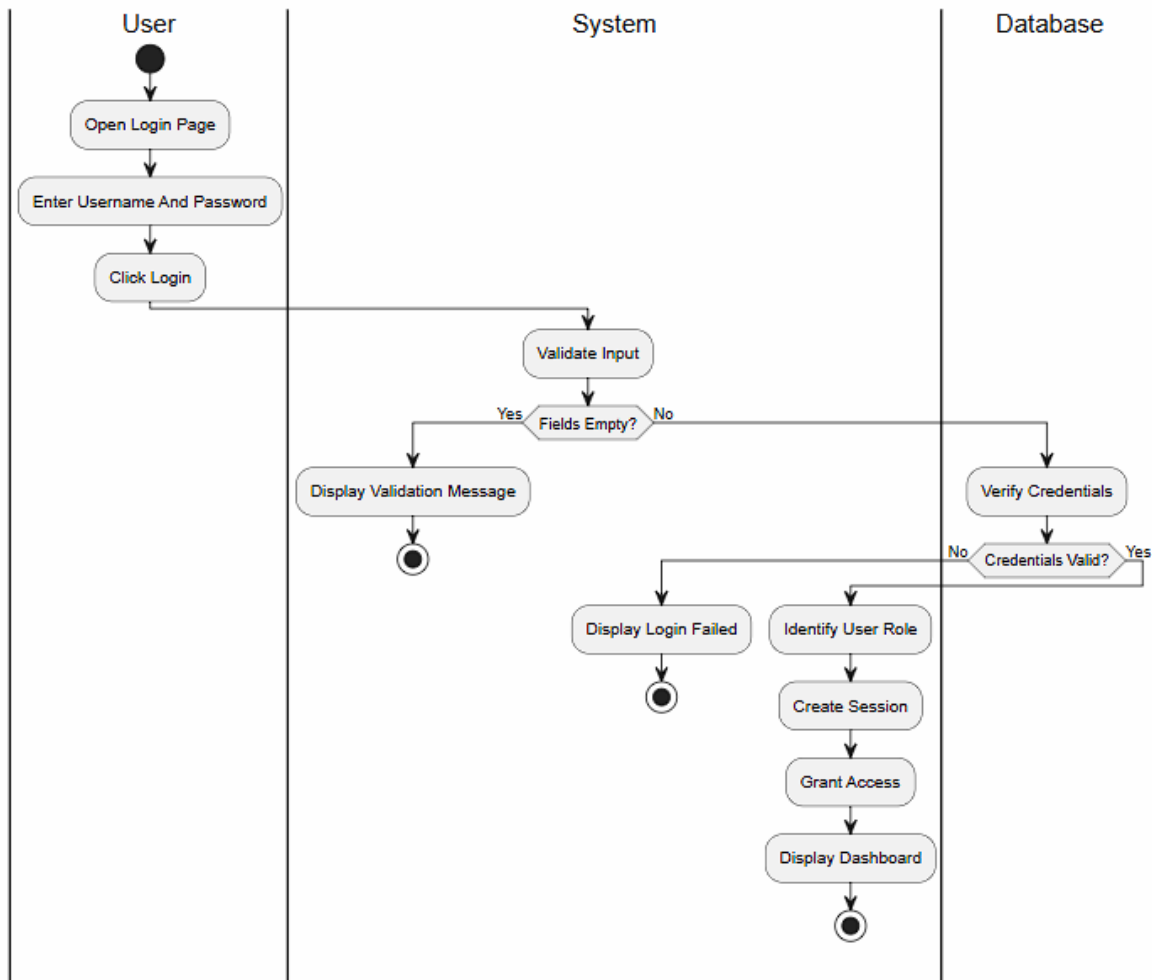
3.1.3 State Diagram



Hình 7: State diagram

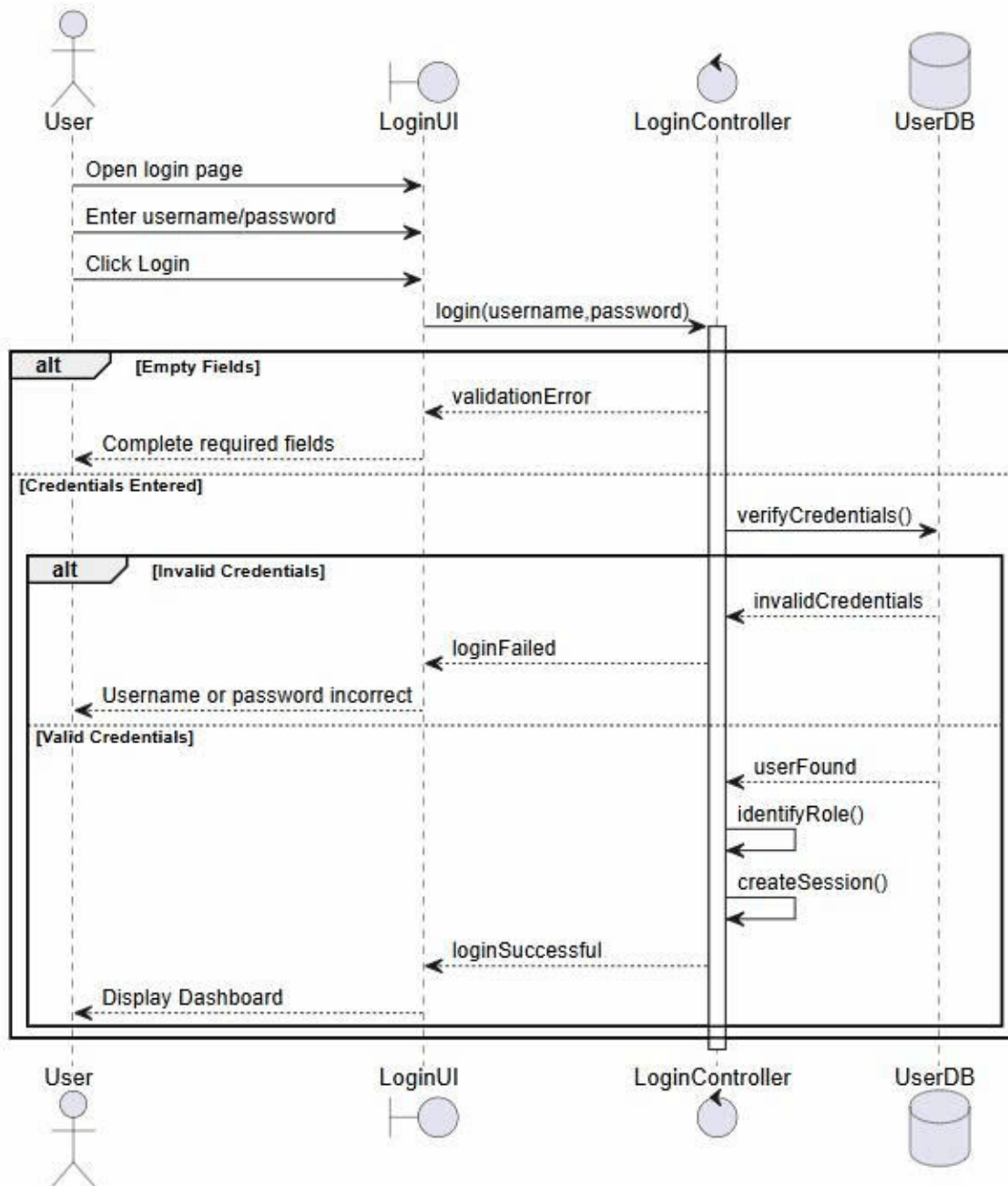
3.2 Login

3.2.1 Activity Diagram



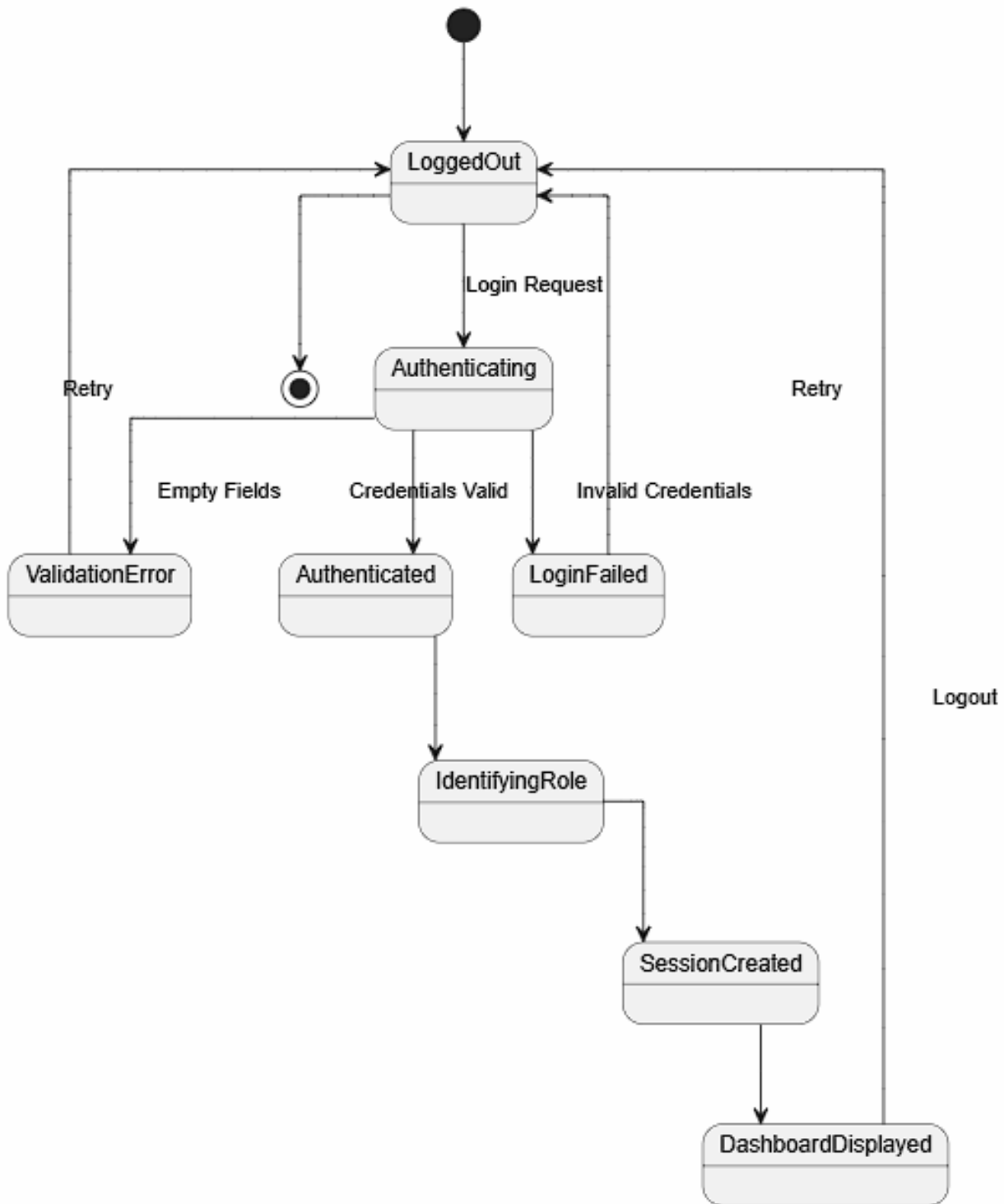
Hình 8: Activity diagram

3.2.2 Sequence Diagram



Hình 9: Sequence Diagram

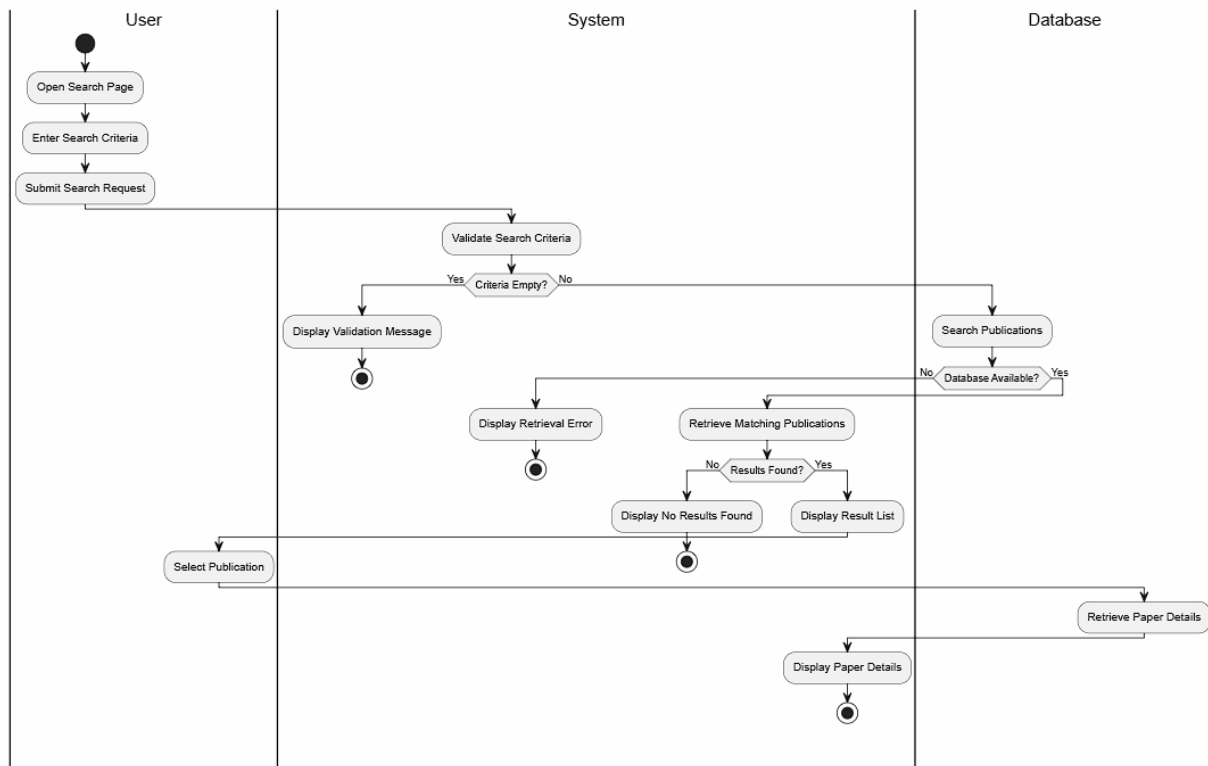
3.2.3 State Diagram



Hình 10: State diagram

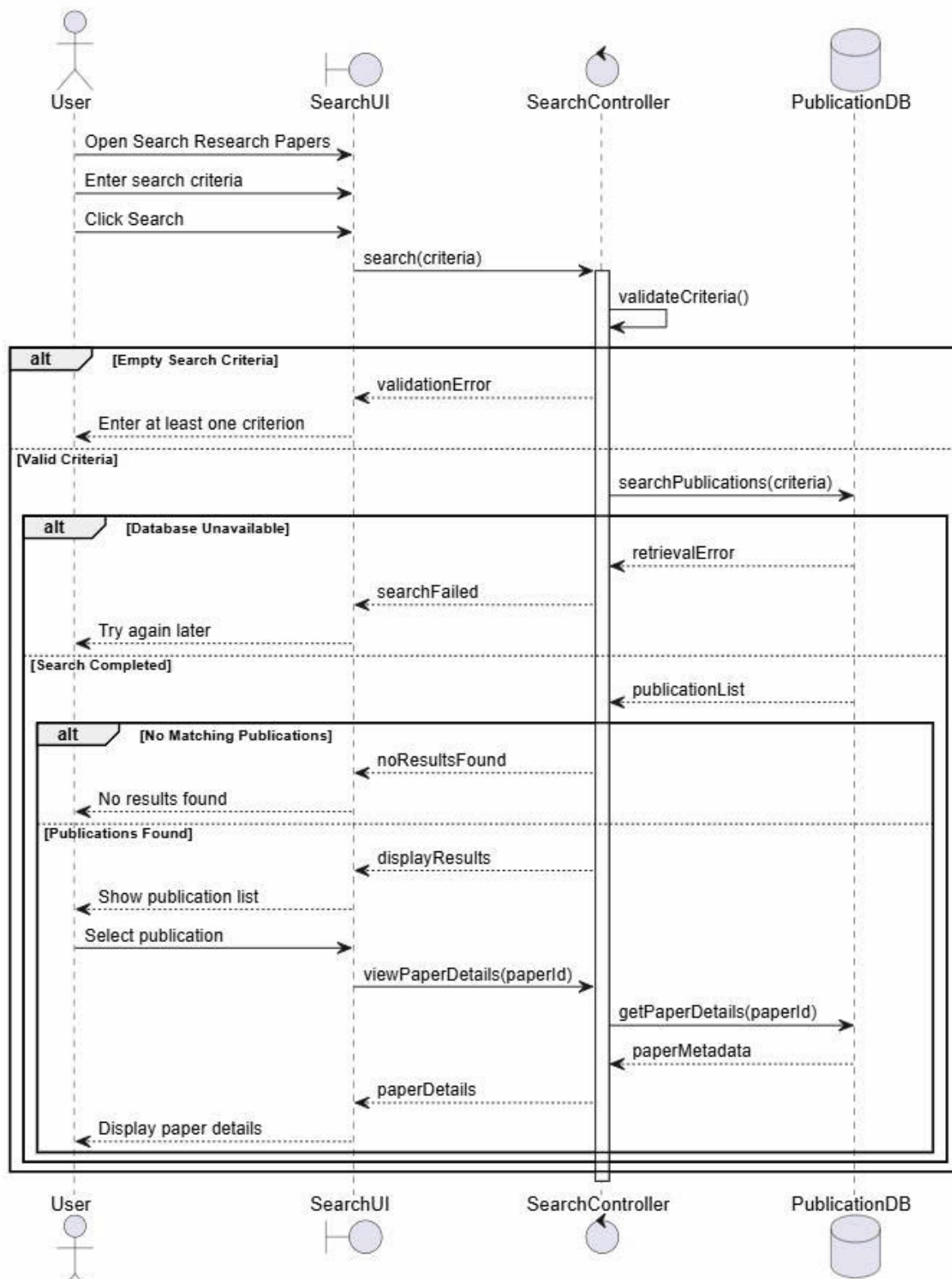
3.3 Search Research Papers

3.3.1 Activity Diagram



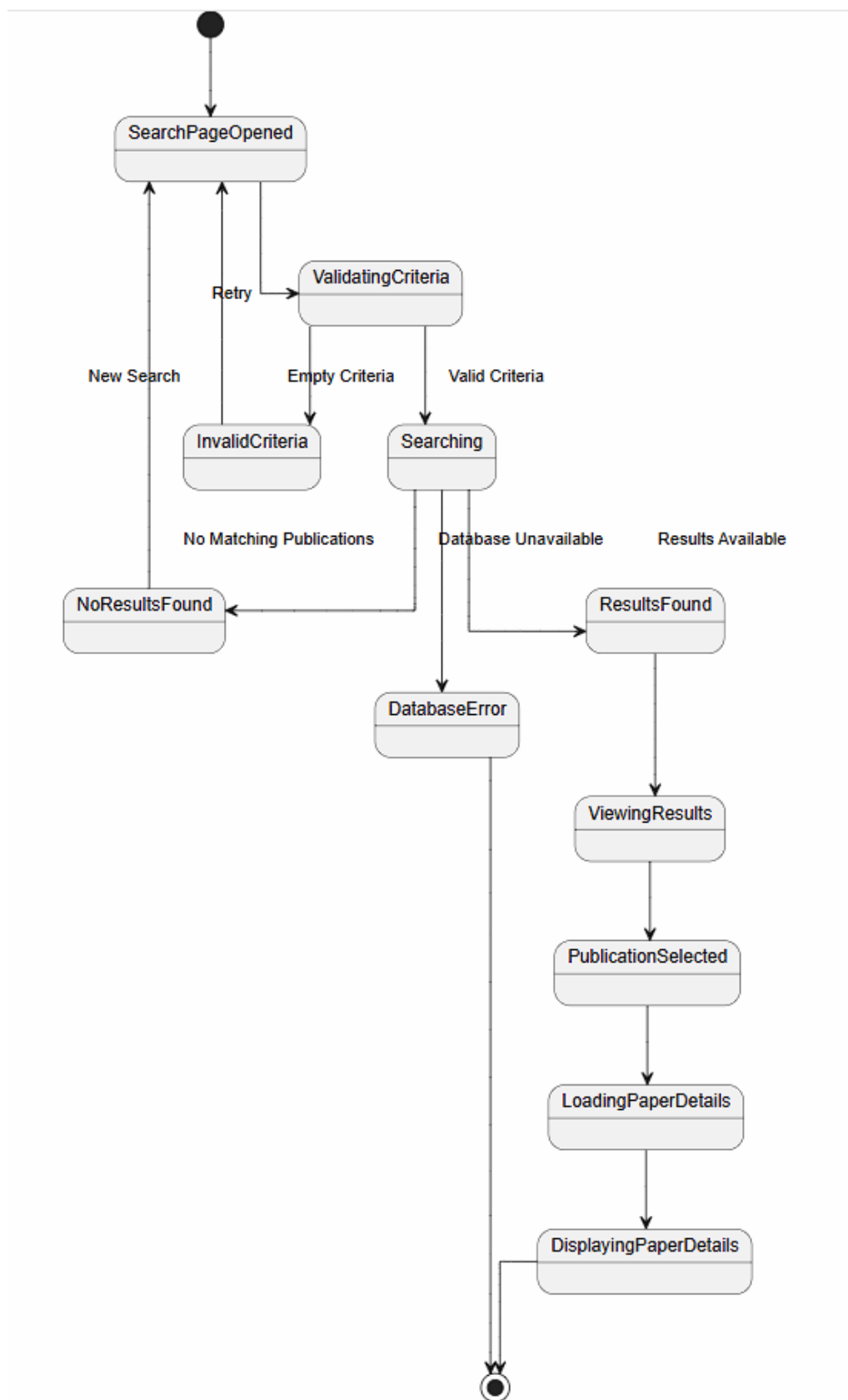
Hình 11: Activity diagram

3.3.2 Sequence Diagram



Hình 12: Sequence Diagram

3.3.3 State Diagram



Hình 13: State diagram

3.4 View Paper Details

3.4.1 Activity Diagram

3.4.2 Sequence Diagram

3.4.3 State Diagram

3.5 Save Bookmarks

3.5.1 Activity Diagram

3.5.2 Sequence Diagram

3.5.3 State Diagram

3.6 Track Publication Trends

3.6.1 Activity Diagram

3.6.2 Sequence Diagram

3.6.3 State Diagram

3.7 View Dashboard

3.7.1 Activity Diagram

3.7.2 Sequence Diagram

3.7.3 State Diagram

3.8 Generate Reports

3.8.1 Activity Diagram

3.8.2 Sequence Diagram

3.8.3 State Diagram

3.9 View Trending Topics

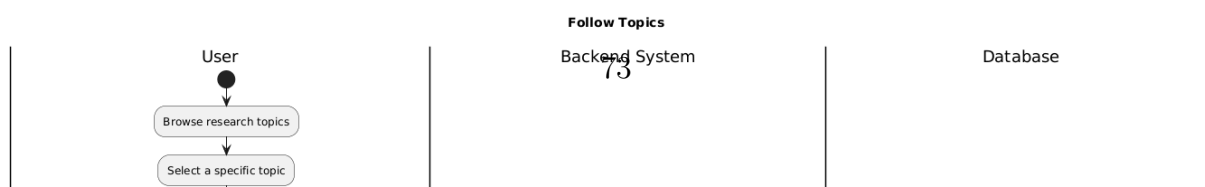
3.9.1 Activity Diagram

3.9.2 Sequence Diagram

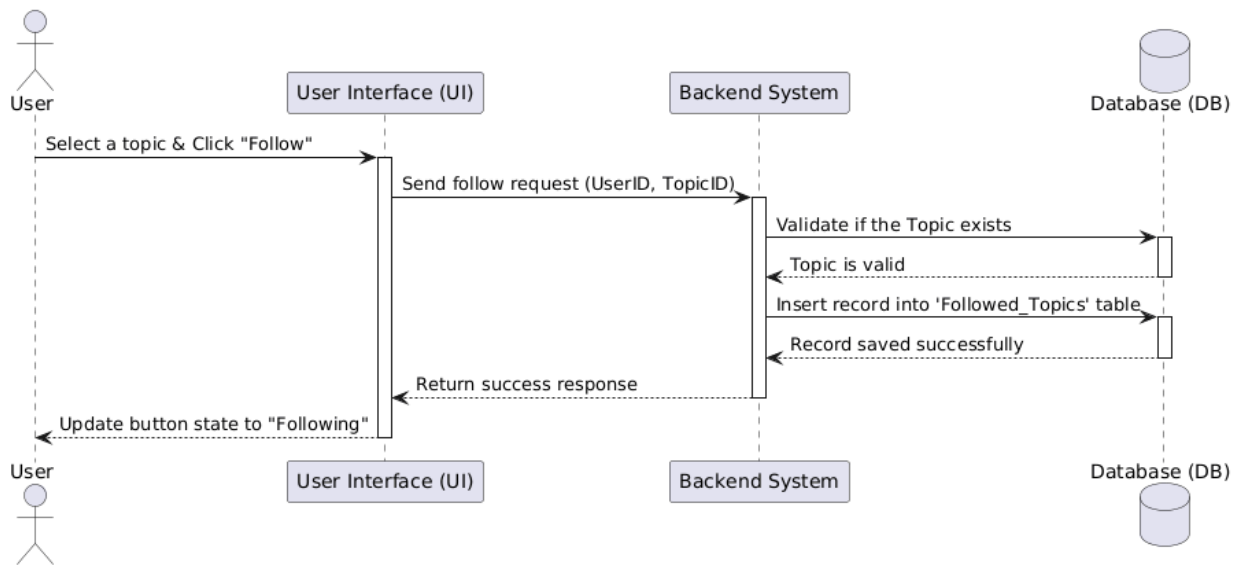
3.9.3 State Diagram

3.10 Follow Topics

3.10.1 Activity Diagram

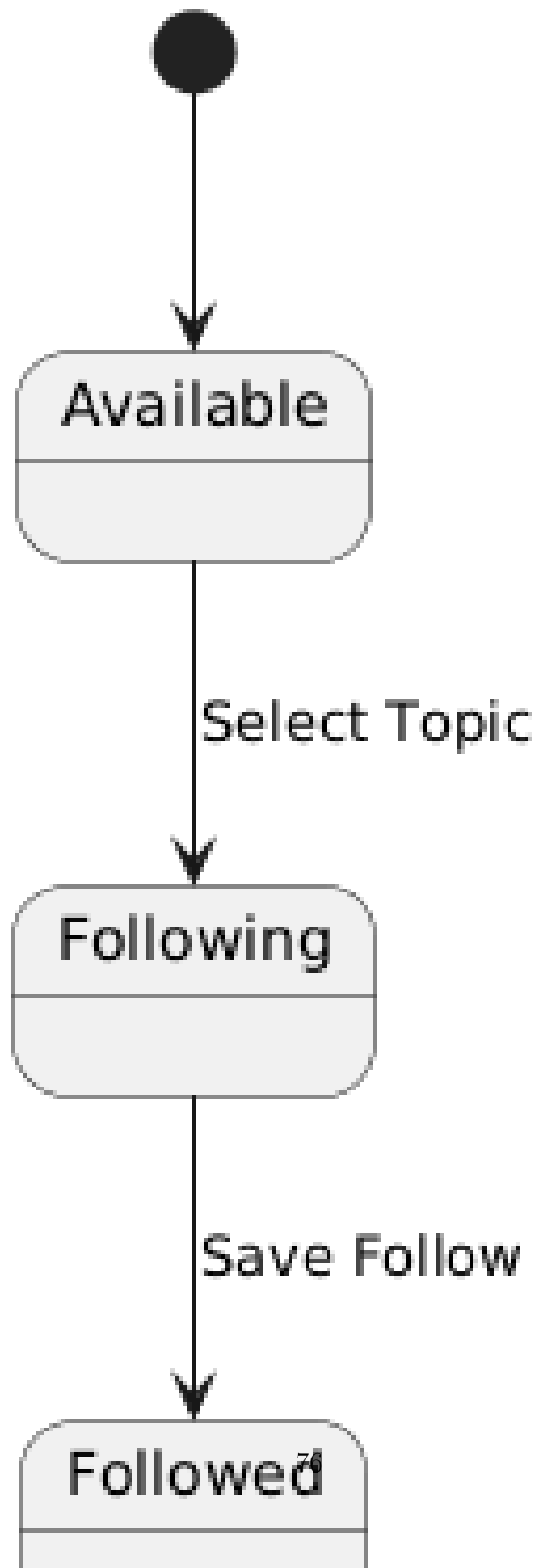


3.10.2 Sequence Diagram



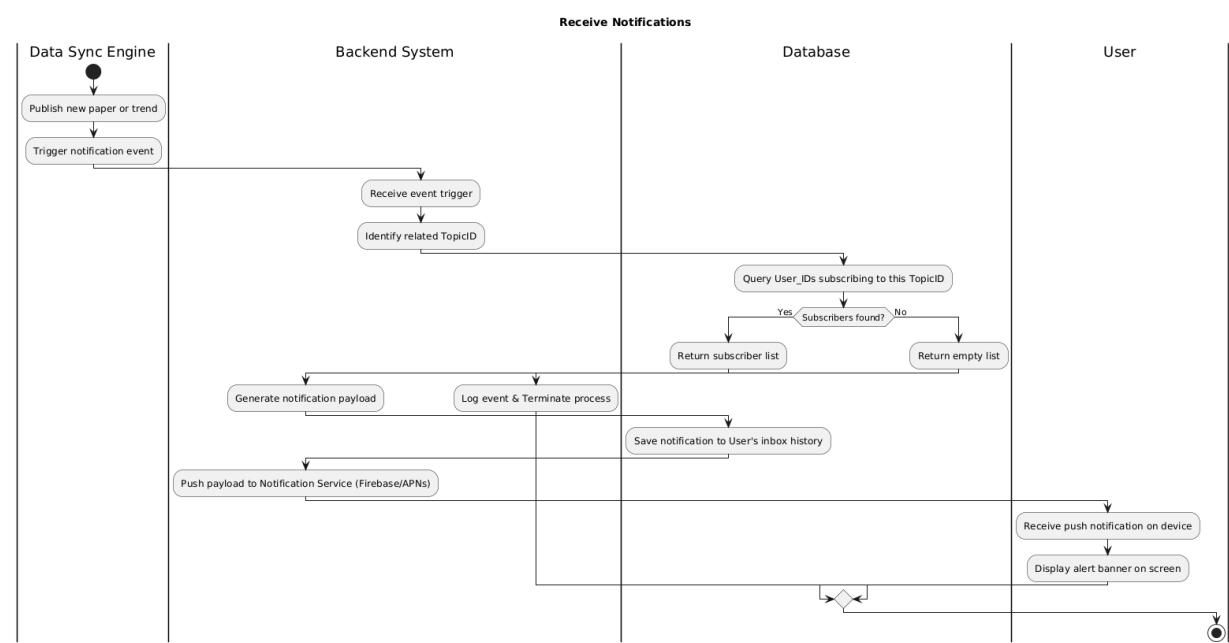
Hình 15: Sequence diagram

State Diagram - Follow Topics



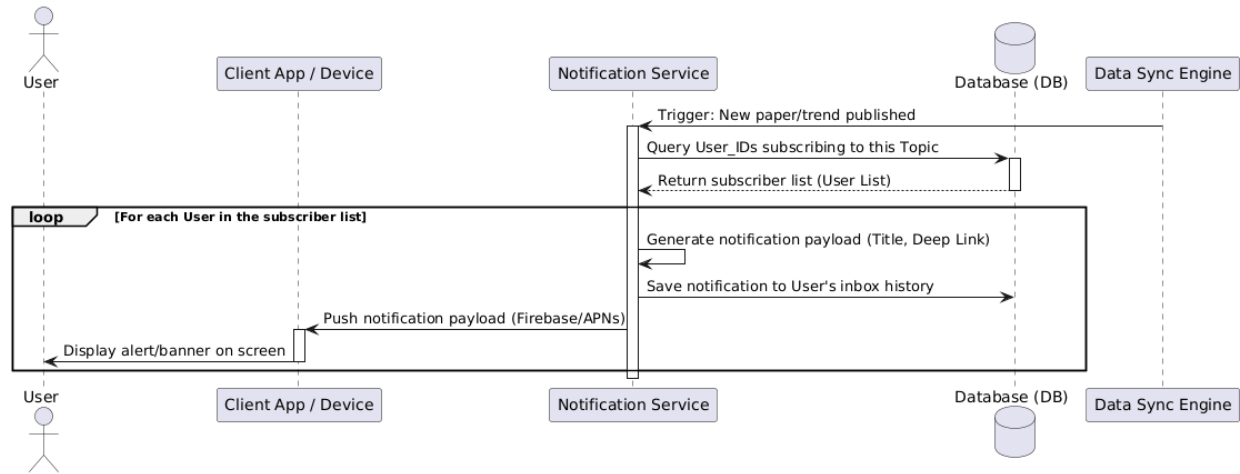
3.11 Receive Notifications

3.11.1 Activity Diagram



Hình 17: Activity diagram

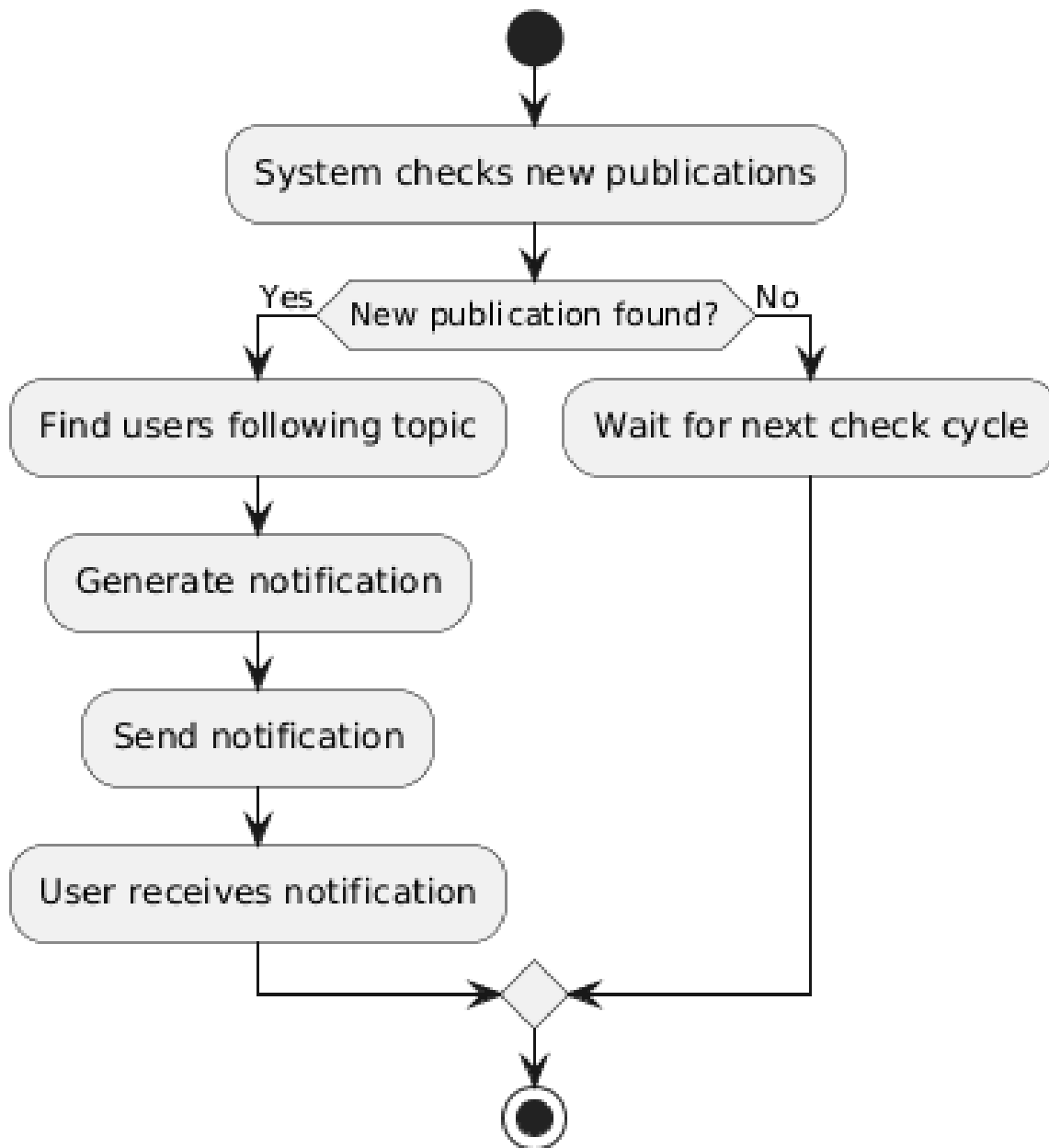
3.11.2 Sequence Diagram



Hình 18: Sequence diagram

3.11.3 State Diagram

State Diagram: 10. Receive Notifications



Hình 19: State diagram

3.12 Manage System

3.12.1 Activity Diagram

3.12.2 Sequence Diagram

3.12.3 State Diagram

3.13 Manage User

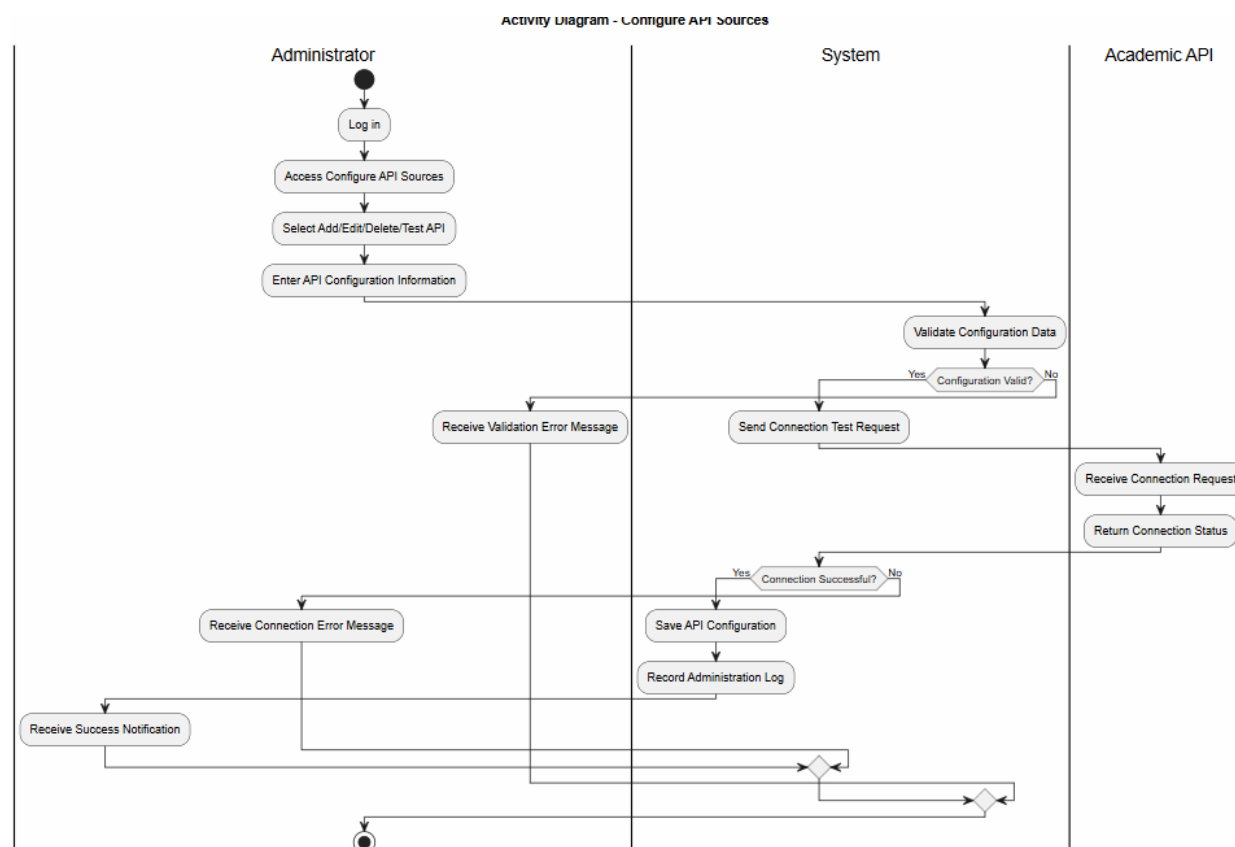
3.13.1 Activity Diagram

3.13.2 Sequence Diagram

3.13.3 State Diagram

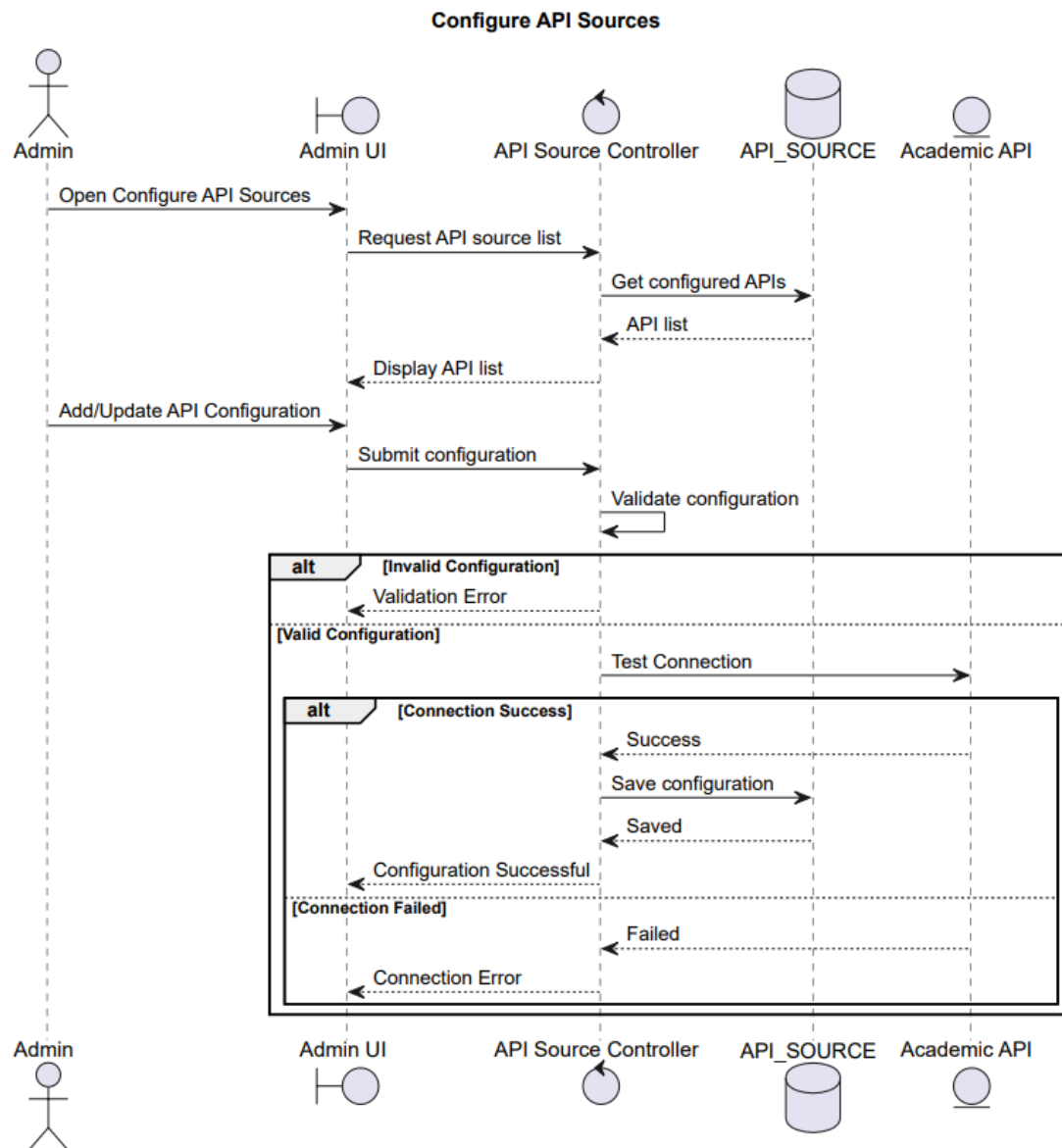
3.14 Configure API Sources

3.14.1 Activity Diagram



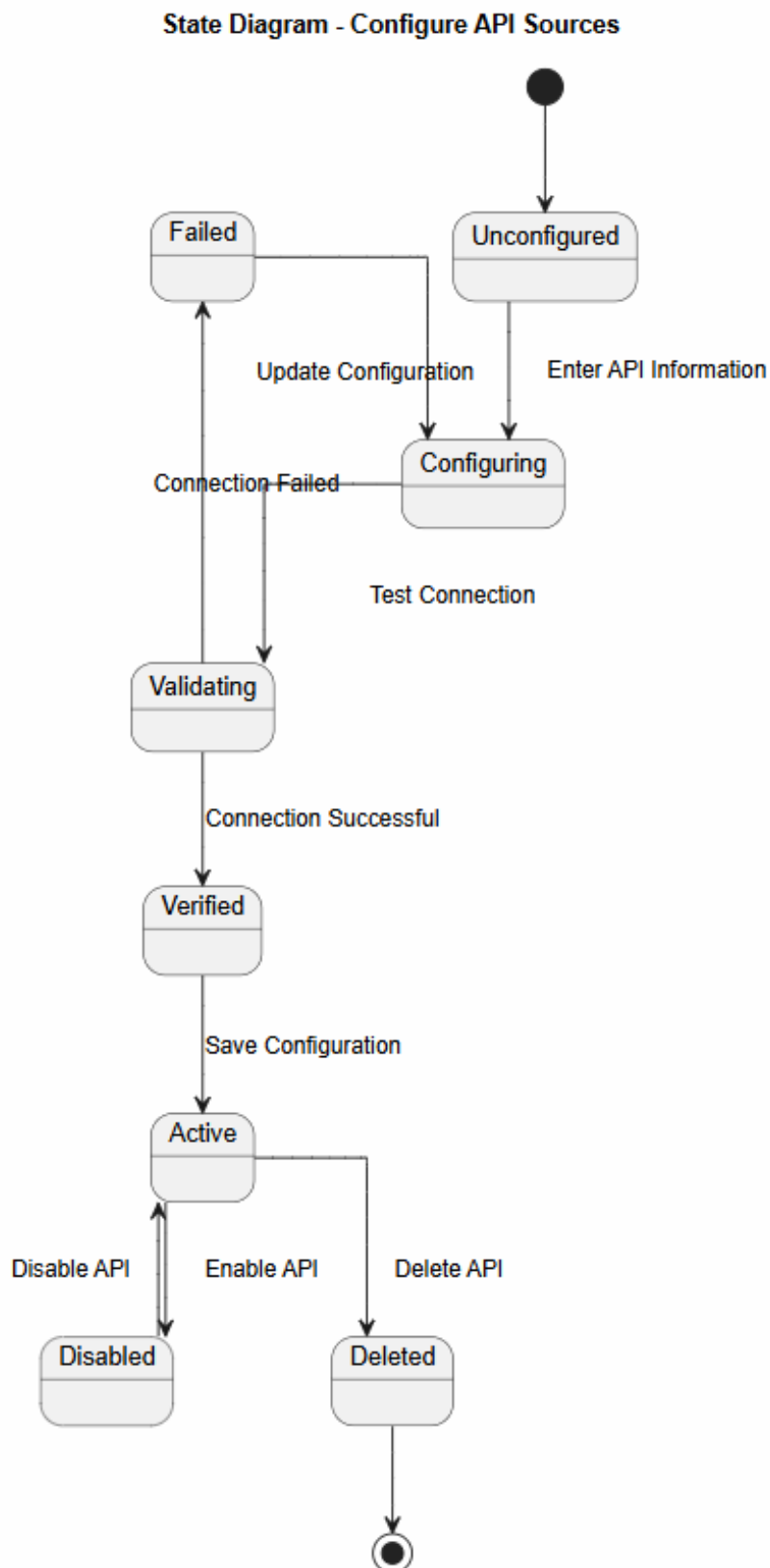
Hình 20: Configure API Sources

3.14.2 Sequence Diagram



Hình 21: Configure API Sources

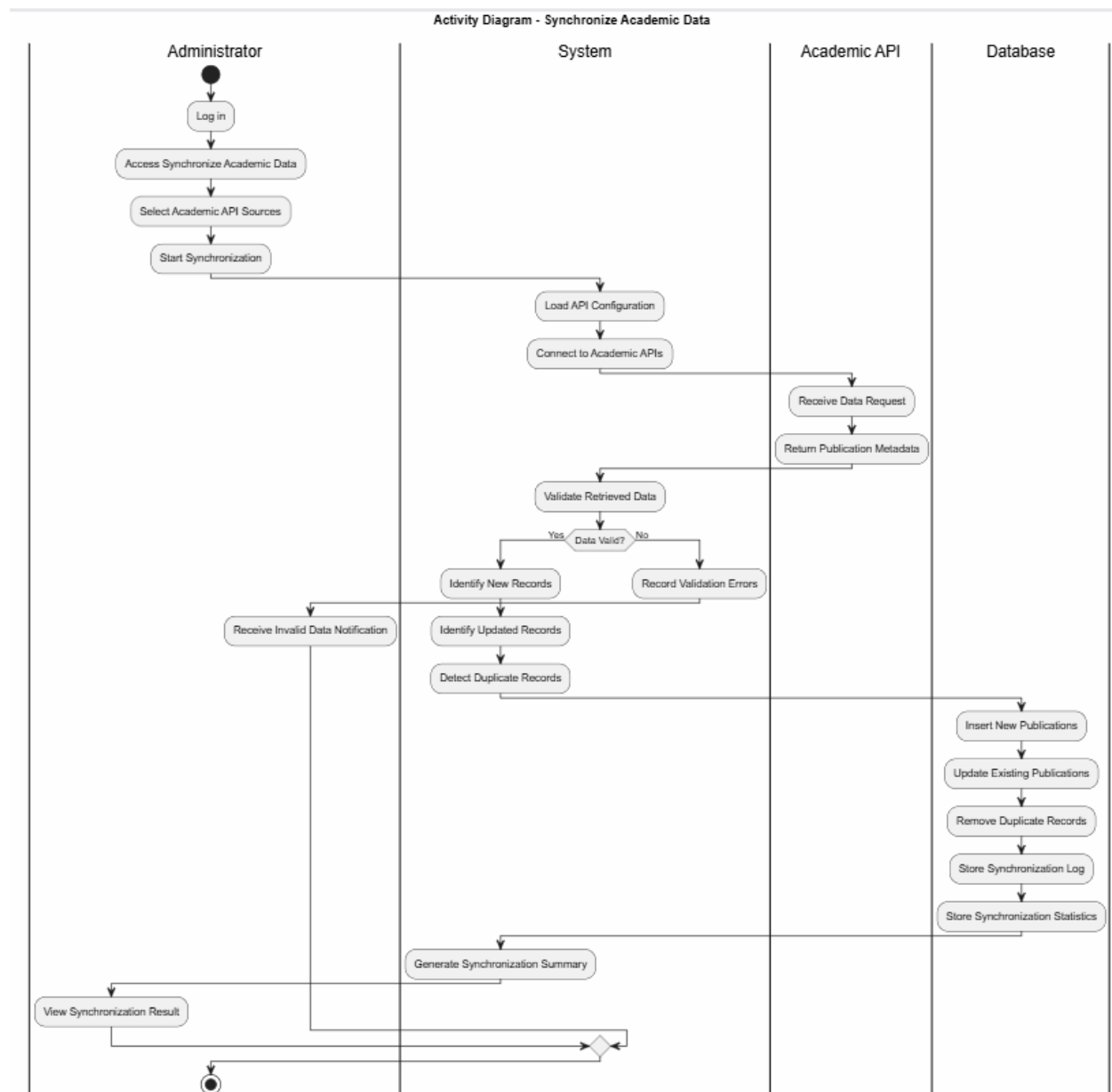
3.14.3 State Diagram



Hình 22: State diagram

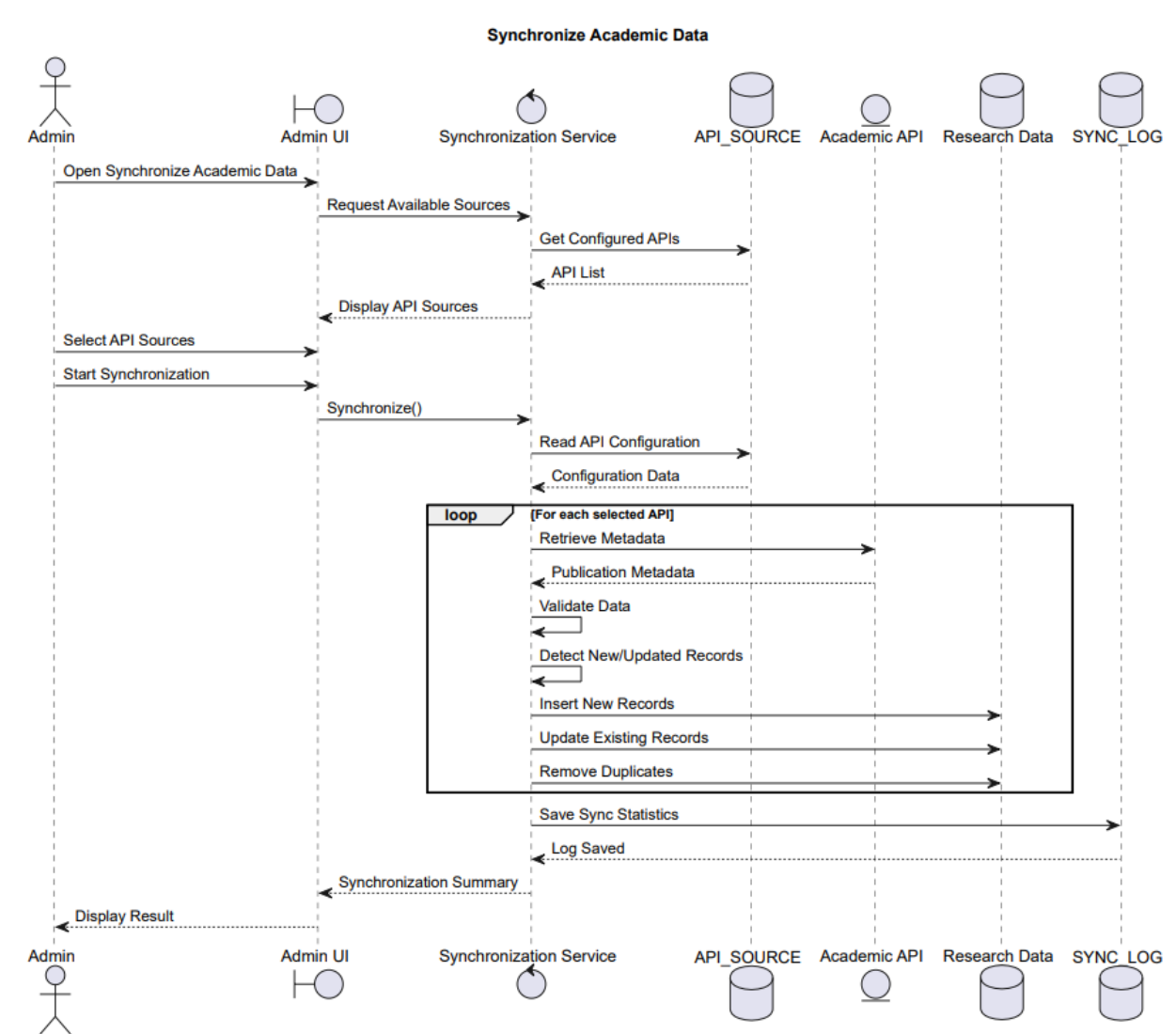
3.15 Sync Academic Data

3.15.1 Activity Diagram



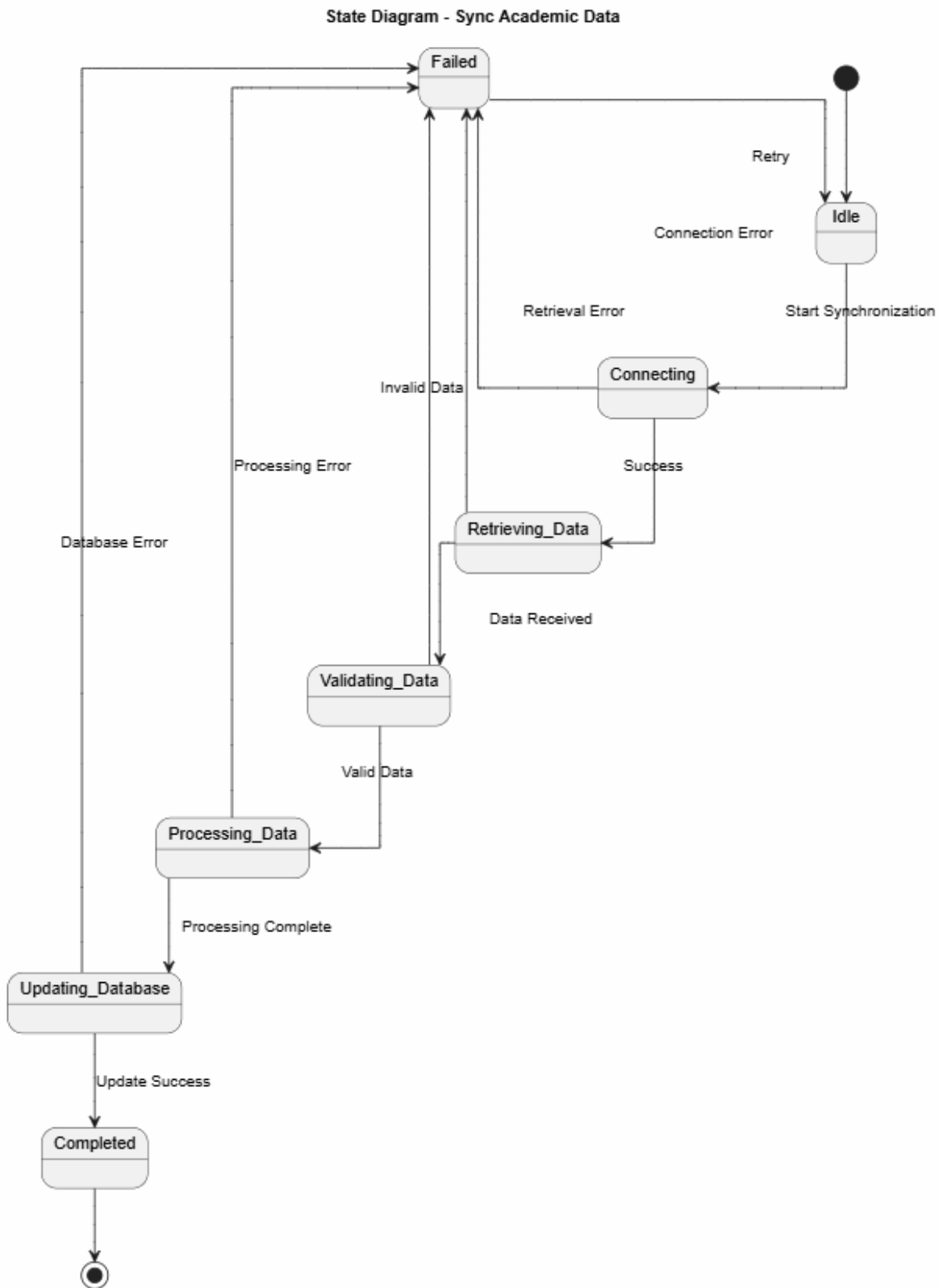
Hình 23: Sync Academic Data

3.15.2 Sequence Diagram



Hình 24: Sync Academic Data

3.15.3 State Diagram



Hình 25: Sync Academic Data